HyTray Invention Disclosure Series

Cascade Impingement Swirl Tray (CIST): A Jet-Fed Swirl-Chamber Distillation Stage with Elliptical Liquid Nozzles

A distillation tray in which vapour–liquid contact occurs in a free swirl-shear zone between opposing jet streams,
decoupled from the gravity-settling flooding limit of conventional decks

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Abstract

A distillation-column tray is disclosed in which vapour–liquid contacting takes place inside an array of short cylindrical swirl chambers, each fed independently by two fluid streams: (i) a liquid jet directed upward through a calibrated elliptical orifice in the chamber floor, powered solely by the liquid static head in the below-tray sump, and (ii) vapour admitted tangentially through a slot in the lower chamber wall, generating a rotating flow field. There is no liquid pool on the tray deck and no perforated flat plate. The swirling vapour shears the ascending liquid jet into a fine annular film and droplet cloud; the rotating two-phase dispersion contacts across the full chamber volume. A fixed-vane crown separator at the chamber top separates the phases centrifugally, passing clean vapour upward and discharging separated liquid outward into a collection ring that drains by sealed conduit to the sump of the next tray below. The elliptical nozzle geometry (aspect ratio $k = a/b = 1.5$–2.5) exploits the axis-switching instability of non-circular jets to produce finer, more uniform droplets than a circular nozzle of equal flow area, increasing interfacial area in the contact zone. The capacity flooding limit is set by centrifugal re-entrainment in the crown, not by gravity settling, predicting a column-diameter reduction of the order of 35–45% relative to a conventional sieve tray at equal tray spacing. Predicted tray efficiency (90% one-pass point efficiency at design conditions) and turndown (6:1 to 8:1, limited by minimum swirl coherence) are estimated from penetration-theory and Stokes-flow models respectively. All performance values are model predictions; they require cold-flow and hot-rig validation before use in detailed design.

Status: All equations and performance estimates in this disclosure are derived from first-principles fluid mechanics and mass-transfer correlations applied to an idealised geometry. No prior tray performance database has been used to calibrate the model. The physical sub-models (Rayleigh–Plateau jet instability, Rankine vortex, Stokes centrifugal collection, penetration-theory mass transfer) are each individually well-established; their application to this combined configuration, and particularly the assumed design parameters (centripetal acceleration, nozzle discharge coefficient, crown collection efficiency), requires experimental confirmation.

Contents

1 Field of the Invention

Vapour–liquid contacting internals for distillation, absorption and stripping columns, specifically a tray-type stage in which vapour–liquid contact is achieved by opposed jet impingement and swirl-shear mixing inside enclosed cylindrical chambers, eliminating the gravity-dependent liquid pool and perforated-plate deck of conventional tray designs.

2 Background and Prior Art

Every conventional distillation tray — whether sieve, valve, or corrugated — shares a common operating principle: vapour passes upward through perforations or valve openings in a horizontal deck, bubbling through a shallow liquid pool maintained by an outlet weir. The vapour–liquid dispersion (froth) on the deck is the contact zone. Phase separation after contact occurs by gravity: liquid drains over the weir to the tray below, and vapour rises clear of the froth.

This configuration imposes three capacity limitations that have not been eliminated simultaneously by any commercial design:

1. **Gravity-settling entrainment limit (Souders-Brown criterion).** Above a critical vapour velocity u_{SB}, liquid droplets entrained from the froth surface cannot settle against the rising vapour stream. This limit depends on tray spacing and system properties but not on any tray geometry parameter under the designer's control beyond the tray-to-tray height.
2. **Liquid weep limit.** Below a minimum vapour velocity, the upward kinetic pressure of the vapour is insufficient to support the liquid head above the perforations and liquid "weeps" downward through the holes, bypassing contact on the tray above. This sets the lower end of the operating range and limits turndown to roughly 2:1–3:1 for conventional sieve and fixed-valve decks.
3. **Downcomer-backup limit.** Liquid from the active deck must descend to the tray below through downcomers, which require a fraction (typically 15–25%) of the column cross-section for their area. At high liquid rates, the downcomer backup can reach the tray above, flooding the column independently of the vapour rate.

Spray-tray and jet-tray concepts have partially addressed the entrainment limit by using mechanical energy (liquid spray) for contact rather than froth aeration, but they typically produce low tray efficiency (short droplet residence time, no multi-pass liquid path) and poor turndown (spray quality collapses at low liquid rate). Centrifugal contactors (cyclone trays, swirl-tube stages) have addressed the entrainment limit by substituting centrifugal for gravitational phase separation, but known designs retain a liquid pool on the deck as the liquid feed to each contactor, re-introducing both the weep limit and the deck-pool entrainment behaviour at the deck surface between contactors.

The CIST eliminates the deck pool entirely. Liquid is fed to each chamber as a coherent jet from below, metered by nozzle hydraulics rather than a weir and pool. Vapour is fed tangentially into the chamber rather than axially through a perforation. Both the contact zone and the phase-separation zone are inside the chamber, making the tray-level flooding limit a function of the swirl field strength, not gravity.

3 Objects of the Invention

1. To eliminate the deck liquid pool, replacing it with a controlled liquid jet fed from a below-tray sump, removing the weep-driven minimum vapour-rate constraint of perforated-plate decks.
2. To replace gravity-settling phase separation with centrifugal separation inside each contact chamber, raising the effective capacity factor by a factor of the order of $\sqrt{N_g}$, where $N_g = a_c/g$ is the centrifugal gravity ratio.
3. To use elliptical rather than circular liquid-jet nozzles to exploit axis-switching jet instability, producing finer and more uniform droplets at the same volumetric flow rate, thereby increasing interfacial area per unit chamber volume.
4. To achieve tray efficiency above 80% one-pass point efficiency by ensuring that the swirling vapour–liquid dispersion contacts across the full chamber height before phase separation at the crown.
5. To provide a stable operating range of at least 6:1 turndown from the lower limit (set by minimum swirl for centrifugal separation) to the upper limit (set by crown re-entrainment).
6. To reduce tray pressure drop relative to conventional decks by eliminating the aerated-liquid-pool contribution βh_{cl} to the pressure balance, replacing it with a smaller vapour-momentum loss through the swirl chamber.
7. To provide a fully cartridge-maintainable stage: all swirl chambers and nozzle assemblies removable through the column manway without shell welding, and all wetted surfaces accessible for inspection.

4 Summary of the Invention

The CIST deck (Figure 1) comprises a horizontal inter-tray plate carrying an array of cylindrical contact chambers on a triangular pitch, a below-plate sump that collects descending liquid, and a liquid overflow seal (weir leg) to the sump of the tray below.

Each contact chamber is a vertical cylinder of inside diameter D_c and height $H_c = (1.2\text{–}2.0)D_c$. An elliptical orifice (nozzle) is machined in the chamber floor; liquid in the sump drives a coherent upward jet through this nozzle under the sump's hydrostatic head. A tangential vapour inlet slot in the lower chamber wall directs the vapour stream circumferentially, establishing a Rankine-type vortex. The ascending liquid jet is sheared by the rotating vapour into an annular film on the chamber wall and a central droplet cloud. Mass transfer proceeds across both the film and the droplet surfaces as the two-phase dispersion ascends the chamber. A vane crown at the chamber top intercepts the dispersion: the heavier liquid phase is deflected outward by centrifugal force into a sealed collection annulus machined into the inter-tray plate, which drains through a sealed conduit to the sump of the tray below; the lighter vapour phase passes upward through the crown to the tray above. No liquid pool exists on the inter-tray plate; all liquid at any instant is either in the sump, ascending as a jet, in the swirl-chamber dispersion, or draining in the collection ring conduit.

The key distinguishing features from all prior vapour–liquid tray art are:

- No perforated deck plate and no liquid pool on a horizontal surface.
- Liquid feed by upward jet from a sump, metered by nozzle orifice hydraulics, not by weir overflow.
- Vapour feed tangentially into a closed cylindrical chamber, not axially upward through a hole in a flat plate.

- Elliptical nozzle geometry exploiting jet axis-switching to reduce the Sauter mean droplet diameter without increasing the nozzle pressure drop.
- Flooding limit set by centrifugal separator performance, not by gravity-settling droplet terminal velocity.

5 Brief Description of Figures

Figure 1 Cross-section elevation of a single CIST chamber showing the sump, elliptical liquid-jet nozzle, tangential vapour inlet, swirling contact zone, crown vane separator, and liquid collection ring with drain conduit.

Figure 2 Plan view (top-down) of a CIST tray showing the hex-packed chamber array, inter-chamber plate, liquid collection ring, and downcomers.

Figure 4 Elliptical nozzle detail: (left) geometry showing the equal-area comparison between ellipse and circle; (right) axis-switching sequence showing the cross-section of the jet at successive distances from the nozzle exit.

Figure 3 Six-view assembly drawing of one CIST swirl chamber: plan view (top), front/rear/side elevations, and two isometric perspectives. Shows the spatial relationship between the sump, nozzle, chamber wall, tangential inlet slot, crown disc, and collection-ring drain.

Figure 5 Wavy-deck variant: (a) wave profile cross-section at $\alpha = 27.5$; (b) three-dimensional perspective showing hex chambers at crests and liquid jets at troughs; (c) single-wave nozzle detail showing liquid film on wave faces and vapour path; (d) flat-vs.-wavy comparison table.

Figure 6 Design refinement details: (a) three nozzle-cut options for the chamber floor (single ellipse, three-ring ellipses, four circular holes); (b) three-nozzle impingement cross-section; (c) wavy-deck chamber-at-crest elevation with axis-switching free-flight zone; (d) smooth wall vs. helical guide vane vs. ring orifice baffle.

Figure 7 Curved helical guide vane: (a) three-dimensional view, colour-coded by local N_g; (b) helix angle $\theta(z)$ from 20° to 70°; (c) centripetal acceleration $N_g(z)$ from 184.7 g to 3.24 g on log scale with effective mean 42.4 g; (d) local cut diameter $d_{50}(z)$ profile, overall $d_{50} = 10.8\,\mu\text{m}$.

Figure 8 Inner-wall enhancement features: (a) elevation cross-section showing saddle-cut base, drainage slots, spiral grooves, liquid film, nozzles, and cross-chevron crown; (b) plan view of cylinder base with saddle-depth contour, slot positions and 120° nozzle layout; (c) feature assessment table; (d) derived film thickness, film Reynolds number, groove enhancement factor and drainage-slot sizing.

Figure 9 Numerical comparison of the CIST (curved helical vane, three elliptical nozzles at 120°) against Sulzer UFM and SVG commercial trays across twelve documented operating cases on a common 9-ft, 2-pass, 24-in-spacing column: (a) per-tray pressure drop; (b) equivalent CIST column diameter at matched throughput; (c) tray efficiency on an O'Connell basis; (d) capacity utilisation and turndown.

6 Detailed Description

6.1 Overall Architecture and Flow Paths

Fig. 1 — CIST Chamber: Cross-Section Elevation

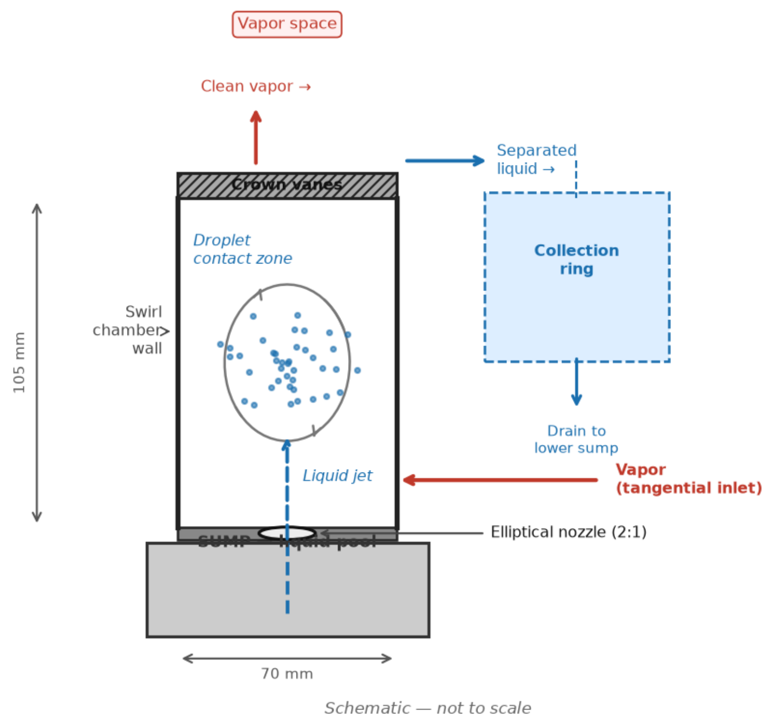

Figure 1: CIST chamber cross-section elevation. Liquid (blue) rises as a jet from the sump through the elliptical nozzle; vapour (red) enters tangentially and rotates in the chamber; the droplet dispersion contacts across the chamber height; the crown vane separates phases, discharging vapour upward and liquid outward to the collection ring.

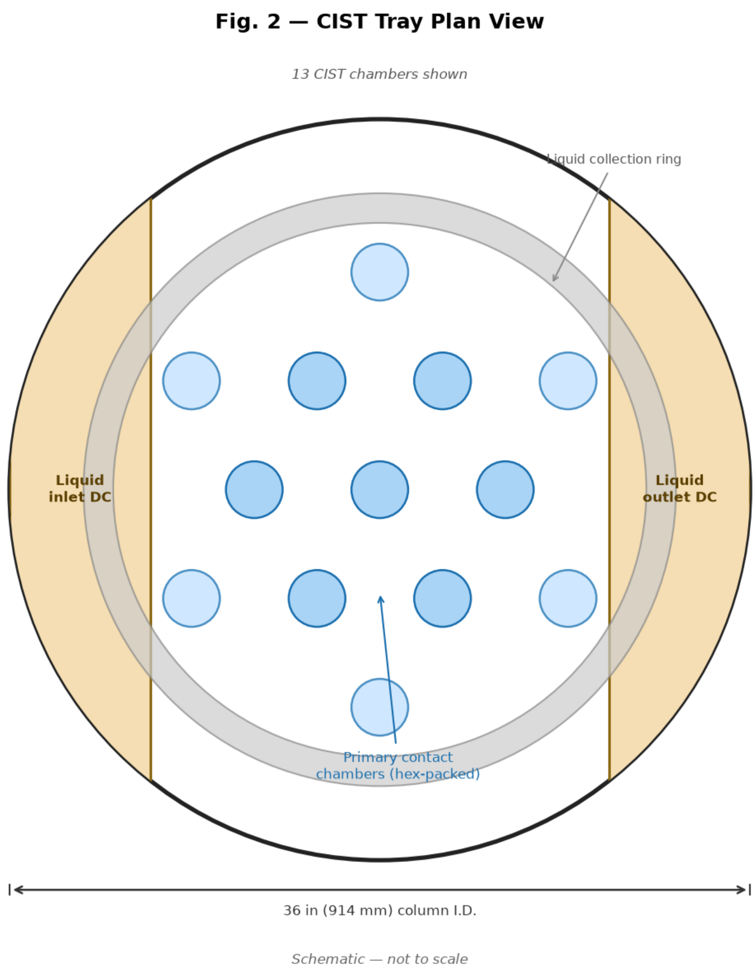

Figure 2: CIST tray plan view. Hex-packed chambers (circles) fill the active area; downcomers (shaded rectangles) occupy the remaining cross-section. Liquid collection rings surround each chamber and connect to sealed drain conduits.

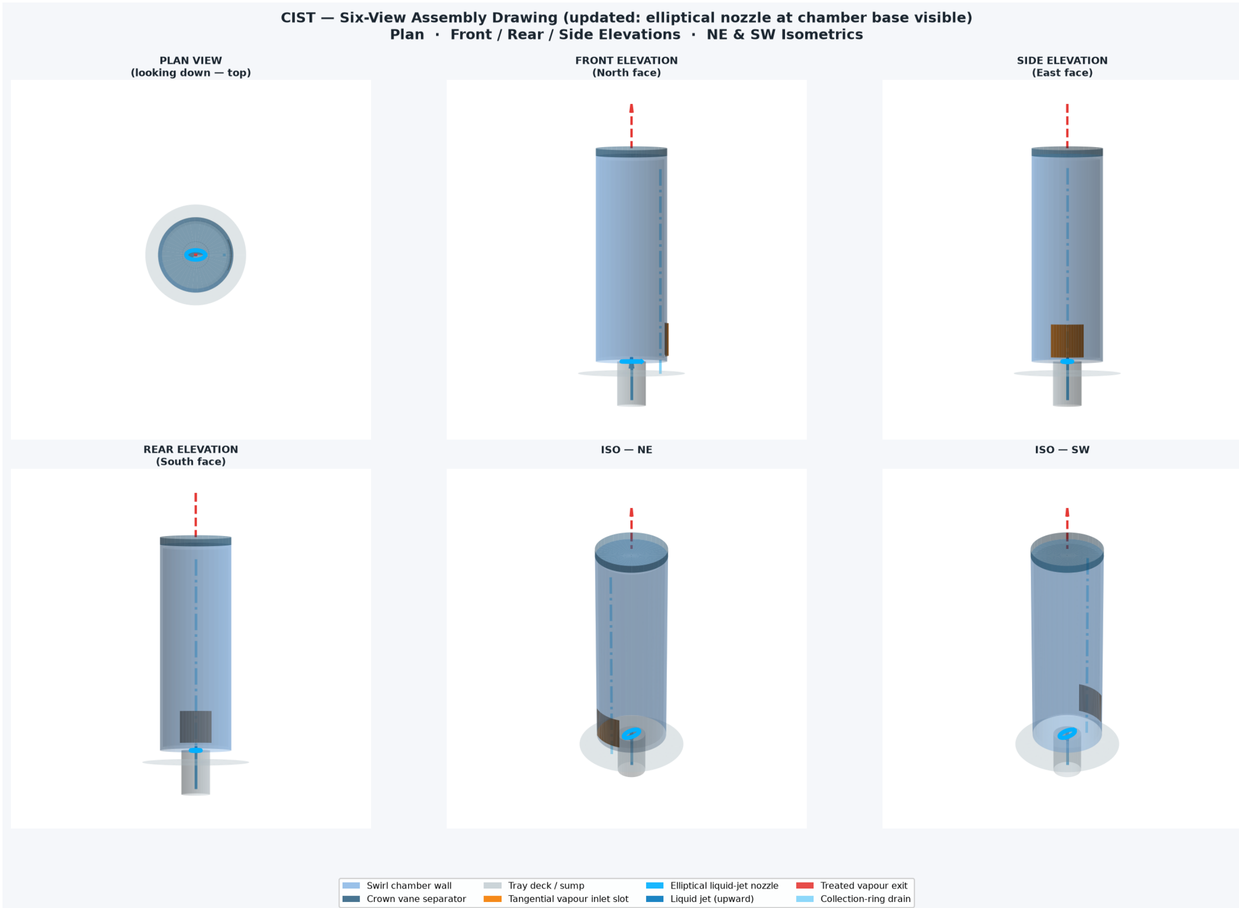

Figure 3: Six-view assembly drawing of one CIST swirl chamber. **Top row, left to right:** plan view (camera directly above), front elevation (North face), side elevation (East face). **Bottom row:** rear elevation (South face), NE isometric, SW isometric. The orange arc on the cylinder surface represents the tangential vapour inlet slot; the upward blue arrow is the liquid jet; the dashed red arrow is the vapour exit above the crown. The dot-dash line on the chamber wall represents the collection-ring drain conduit.

The CIST tray is bounded above and below by two horizontal inter-tray plates separated by the tray spacing H_S. The upper plate carries the chamber array; the lower plate forms the floor of the sump. Flow paths are:

Vapour path Enters the inter-tray space from the tray below, splits into the tangential inlet slots of each chamber, spirals up through the contact zone, and exits clean through the crown vanes.

Liquid path Descends from the tray above through a sealed overflow leg into the below-plate sump; is drawn upward as individual jets through the elliptical nozzles by hydrostatic head; contacts vapour in the swirl zone; is separated at the crown; and drains via the collection ring conduit back down to the sump of the tray below.

Downcomers (not shown in Figure 1) occupy approximately 15% of the column cross-section at the tray periphery, serving as the sealed liquid-overflow legs between sump levels. The chamber array occupies the remaining 85% of the column cross-section; the hex-packed chamber tubes themselves occupy approximately 35% of that area ($f_{\text{chamber}} = 0.35$, from $\pi/(2\sqrt{3}) \approx 0.907$ hex packing times a chosen chamber-to-pitch ratio $(D_c/p)^2$ where p is the centre-to-centre pitch).

6.2 Liquid Sump and Overflow Seal

The sump is a shallow pan of height $h_s = 40–80\,\mathrm{mm}$ formed between the inter-tray plate and the lower deck. Liquid accumulates in the sump to a steady level H_L above the nozzle orifice plane; this head drives the liquid jets. A sealed overflow leg (a pipe sealed against vapour short-circuit) maintains H_L within 10–20 mm across the operating range by balancing the liquid input rate (from the tray above) with the jet discharge rate summed over all N_c chambers. Because all jets share one sump, per-chamber liquid flow self-equalizes under the common head — no individual metering adjustment is required.

6.3 Elliptical Liquid-Jet Nozzle

6.3.1 Geometry

The nozzle is a sharp-edged elliptical orifice in the chamber floor plate, with semi-major axis a (horizontal) and semi-minor axis b (horizontal, perpendicular), aspect ratio $k = a/b \geq 1$.

Fig. 3 — Elliptical Nozzle: Geometry and Axis-Switching Phenomenon

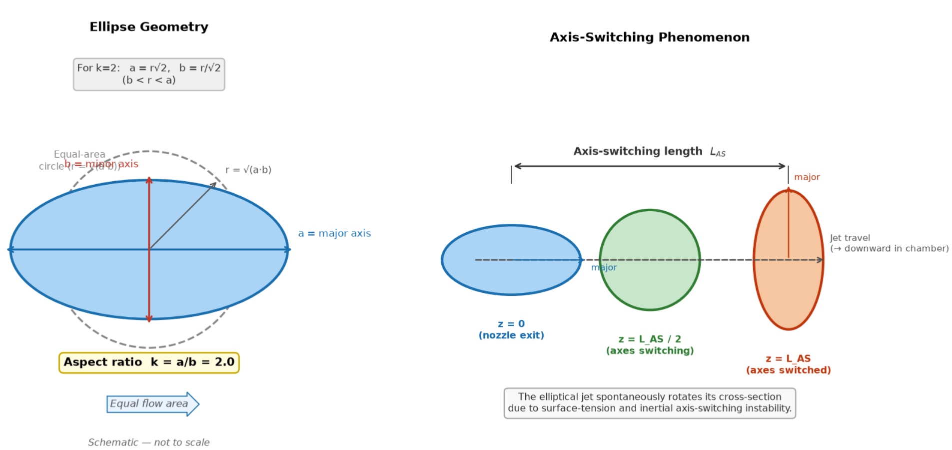

Figure 4: Elliptical nozzle geometry (left) and axis-switching sequence (right). The dashed circle shows the equal-area circular orifice; the ellipse minor axis b is smaller by a factor $1/\sqrt{k}$, producing finer minimum droplets.

The nozzle open area:

$$A_n = \pi ab = \pi k b^2 \tag{1}$$

For equal area with a circular nozzle of radius r_0: $\pi ab = \pi r_0^2$, giving $a = r_0\sqrt{k}$ and $b = r_0/\sqrt{k}$. Thus the minor axis is smaller than the equivalent circle radius by $1/\sqrt{k}$, and the major axis is larger by $\sqrt{k}$.

The hydraulic diameter, which governs frictional losses and the discharge coefficient, is:

$$D_h = \frac{4A_n}{P_n} \tag{2}$$

where P_n is the nozzle perimeter. For an ellipse no exact closed form exists; the Ramanujan (1914) approximation is accurate to better than 0.02%:

$$P_n \approx \pi\left[3(a+b) - \sqrt{(3a+b)(a+3b)}\right] \tag{3}$$

For $k = 2$ (design preferred): $a = r_0\sqrt{2}$, $b = r_0/\sqrt{2}$; $P_n \approx \pi\left[3 \cdot 3r_0/\sqrt{2} - \sqrt{5r_0/\sqrt{2} \cdot 7r_0/\sqrt{2}}\right] = \pi r_0[9/\sqrt{2} - \sqrt{35}/\sqrt{2}] = \pi r_0(9 - \sqrt{35})/\sqrt{2} \approx \pi r_0 \cdot 2.180$. Comparing with the circle ($P_{\text{circle}} = 2\pi r_0$): the 2:1 ellipse perimeter is ≈ 1.09 times the circle perimeter at equal area, confirming that $D_h^{\text{ellipse}} \approx 0.917\, D_h^{\text{circle}}$.

6.3.2 Orifice Hydraulics: Bernoulli Derivation

The jet velocity is derived from the mechanical-energy equation (Bernoulli) for steady, incompressible, inviscid flow between the sump surface (station 1) and the nozzle vena contracta (station 2):

$$\frac{P_1}{\rho_L g} + \frac{u_1^2}{2g} + z_1 = \frac{P_2}{\rho_L g} + \frac{u_2^2}{2g} + z_2 \tag{4}$$

The sump surface is large; continuity gives $u_1 \approx 0$. Both stations communicate with the same vapour space so $P_1 = P_2$. Taking the nozzle plane as datum ($z_2 = 0$) and the sump surface at $z_1 = H_L$:

$$H_L = \frac{u_{\text{ideal}}^2}{2g} \quad \Rightarrow \quad u_{\text{ideal}} = \sqrt{2gH_L} \tag{5}$$

The real exit velocity is reduced by the vena contracta effect at the sharp-edged orifice. Defining the discharge coefficient C_d as the ratio of actual volumetric flow to the ideal:

$$u_{\text{jet}} = C_d\sqrt{2gH_L} \quad ; \quad Q_{\text{nozzle}} = C_d A_n \sqrt{2gH_L} \tag{6}$$

For a sharp-edged elliptical orifice at Reynolds number $Re = \rho_L u_{\text{jet}} D_h/\mu_L > 10^4$, the discharge coefficient is well approximated by a correction to the circular-orifice value $C_{d,\circ} \approx 0.611$:

$$C_d = C_{d,\circ}\left(\frac{D_h^{\text{ellipse}}}{D_h^{\text{circle}}}\right)^{0.04} \tag{7}$$

This correction, derived from the empirical dependence of C_d on the orifice perimeter-to-area ratio (Lichtarowicz *et al.*, 1965), gives $C_d \approx 0.607$ for $k = 1.5$, $C_d \approx 0.601$ for $k = 2.0$, and $C_d \approx 0.593$ for $k = 2.5$. The pressure drop across the nozzle at the design flow rate Q_{nozzle} is:

$$\Delta P_{\text{nozzle}} = \rho_L g H_L = \frac{\rho_L}{2}\left(\frac{Q_{\text{nozzle}}}{C_d A_n}\right)^2 \tag{8}$$

which is identically equal to the hydrostatic head powering the jet — all of the available head is converted to jet kinetic energy (minus the contraction loss already absorbed into C_d).

6.4 Jet Formation: Weber Number and Minimum-Flow Condition

A coherent jet forms only above a minimum Weber number based on the minor axis $2b$ (the characteristic length of the ellipse that sets the fastest instability):

$$We_L = \frac{\rho_L u_{\text{jet}}^2 \cdot 2b}{\sigma} \tag{9}$$

Below $We_L \approx 8$ the surface tension forces dominate and the liquid exits as a drip rather than a jet; above $We_L \approx 8$ a coherent jet forms and travels through the chamber before breaking up. The minimum sump head for jet formation is therefore:

$$H_{L,\min} = \frac{4b\,\sigma}{\rho_L g C_d^2 (2b)} = \frac{2\sigma}{\rho_L g C_d^2} \times \frac{We_{L,\min}}{2b} \tag{10}$$

Substituting $We_{L,\min} = 8$:

$$H_{L,\min} = \frac{8\sigma}{\rho_L C_d^2 \cdot 2b \cdot g} = \frac{4\sigma}{\rho_L g C_d^2 b} \tag{11}$$

For a light-hydrocarbon system ($\sigma = 10\,\text{dyn/cm} = 0.010\,\text{N/m}$, $\rho_L = 500\,\text{kg/m}^3$) and $b = 3\,\text{mm}$, $C_d = 0.60$: $H_{L,\min} = 4 \times 0.010/(500 \times 9.81 \times 0.36 \times 0.003) = 0.040/(5.30) \approx 7.5\,\text{mm}$. A sump head of 40–80 mm (design operating range) keeps We_L well above the drip-to-jet transition throughout the turndown range.

6.5 Jet Breakup Physics: Rayleigh Instability and Axis-Switching

6.5.1 Rayleigh–Plateau instability for a circular jet

A cylindrical jet of radius r_0 and surface tension σ is subject to capillary (Rayleigh-Plateau) instability. A small sinusoidal perturbation of wavenumber $k_r = 2\pi/\lambda$ on the jet surface grows at a rate ω given by (Rayleigh 1878):

$$\omega^2 = \frac{\sigma}{\rho_L r_0^3} \frac{I_1(k_r r_0)}{I_0(k_r r_0)} k_r r_0 \big(1 - k_r^2 r_0^2\big) \tag{12}$$

where I_0, I_1 are modified Bessel functions of the first kind. The disturbance grows only for $k_r r_0 < 1$, i.e. for wavelengths $\lambda > 2\pi r_0$ (the Rayleigh criterion: the wavelength must exceed the jet circumference). The fastest-growing mode occurs at $k_r r_0 \approx 0.6966$ (obtained by maximising Eq. 12), giving:

$$\lambda_{\text{opt}} = \frac{2\pi r_0}{0.6966} \approx 9.016\, r_0 \tag{13}$$

The breakup length L_b (distance at which the fastest-growing mode reaches amplitude equal to r_0) is proportional to the jet velocity:

$$L_b = u_{\text{jet}}/\omega_{\max} \tag{14}$$

where $\omega_{\max}$ is evaluated at $k_r r_0 = 0.6966$. For typical process conditions $L_b \approx 80$–$200\,\text{mm}$, which must be less than H_c for breakup to occur inside the chamber.

6.5.2 Axis-switching instability for an elliptical jet

An elliptical jet cross-section is not in surface-tension equilibrium. The Young-Laplace pressure excess at a point on the surface with principal radii of curvature R_1 and R_2 is:

$$\Delta P = \sigma \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \tag{15}$$

For an ellipse, the curvature varies continuously around the perimeter. At the tip of the minor axis (highest curvature, radius $\kappa_{\text{tip}} = a^2/b$):

$$\Delta P_{\text{tip}} \approx \frac{\sigma\, a^2}{b^2\, r_0} \tag{16}$$

At the flank of the major axis (lowest curvature, radius $\kappa_{\text{flank}} = b^2/a$):

$$\Delta P_{\text{flank}} \approx \frac{\sigma\, b^2}{a^2\, r_0} \tag{17}$$

The pressure excess $\Delta P_{\text{tip}} > \Delta P_{\text{flank}}$ drives liquid flow from the minor-axis tips toward the major-axis flanks, deforming the cross-section: the minor axis elongates and the major axis contracts. The cross-section oscillates between two orthogonal elliptical orientations — "axis-switching" — at a characteristic spatial period L_{AS}. From dimensional analysis and experimental observation (Kasyap *et al.*, 2009; Mourtazin and Cohen, 2007):

$$L_{\text{AS}} \approx C_{\text{AS}} \sqrt{\frac{\rho_L\, u_{\text{jet}}^2\, a^3}{\sigma}} \tag{18}$$

where $C_{\text{AS}} \approx 2.0$–$3.5$ is an empirical constant that depends weakly on aspect ratio. The axis-switching phenomenon has two consequences for the CIST:

1. **Faster breakup.** The oscillating minor axis generates capillary disturbances at a wavelength shorter than the Rayleigh optimal λ_{opt}. The ellipse breakup occurs at:

$$\lambda_b^{\text{ellipse}} \approx \lambda_{\text{opt}} \times \sqrt{\frac{b}{r_0}} = 9.016\, r_0 \times \frac{1}{\sqrt{k}} \tag{19}$$

   and the breakup length shortens accordingly. For $k = 2$: the ellipse breaks up in $\approx 71\%$ of the circular jet breakup distance.
2. **Finer droplets.** Each axis-switching half-cycle pinches off a ring of droplets at the point of maximum curvature change. The mean droplet diameter from the primary breakup of an elliptical jet at aspect ratio k is estimated as:

$$d_{32}^{\text{ellipse}} \approx 1.89 \times 2b = \frac{3.78\, r_0}{\sqrt{k}} \tag{20}$$

   For $k = 2$: $d_{32} \approx 2.67\, r_0$ versus $d_{32} \approx 3.78\, r_0$ for a circular jet of the same flow area — a 29% reduction in Sauter mean diameter at equal flow rate. Since interfacial area per unit volume $a_d = 6/d_{32}$ (for spherical droplets), this is a 40% increase in interfacial area density.

The minimum droplet diameter produced by the elliptical jet scales with the minor axis:

$$d_{\min} \approx 1.0 \times 2b = \frac{2r_0}{\sqrt{k}} \tag{21}$$

This is smaller than the equivalent circular jet minimum ($d_{\min} = 2r_0$) by a factor $1/\sqrt{k}$. At $k = 2$: $d_{\min}^{\text{ellipse}} = 0.707 d_{\min}^{\text{circle}}$. Finer minimum droplets are harder to separate centrifugally (see Section 6.8), so the swirl field must be proportionally stronger to maintain the same capture efficiency; this is the principal design trade-off of the elliptical nozzle.

6.6 Swirl Chamber: Tangential Vapour Inlet and Rankine Vortex

6.6.1 Rankine vortex model

The tangential vapour inlet creates a swirling flow inside the chamber. For a steady, axisymmetric, incompressible flow in a cylindrical container, the azimuthal velocity profile is modelled as a combined (Rankine) vortex:

$$u_\theta(r) = \begin{cases} \Omega r & r \le r_c \quad \text{(solid-body rotation, inner core)} \\ \dfrac{\Omega r_c^2}{r} & r > r_c \quad \text{(free vortex, outer annulus)} \end{cases} \tag{22}$$

where Ω (rad/s) is the angular velocity at the core radius r_c. The core is formed by viscous coupling near the chamber centreline; typically $r_c \approx 0.3$–$0.5\, R_c$. The centripetal acceleration at radius r:

$$a_c(r) = \frac{u_\theta(r)^2}{r} = \begin{cases} \Omega^2 r & r \le r_c \\ \dfrac{\Omega^2 r_c^4}{r^3} & r > r_c \end{cases} \tag{23}$$

The acceleration at the wall ($r = R_c$) is:

$$a_{c,w} = \frac{u_\theta(R_c)^2}{R_c} \tag{24}$$

6.6.2 Relationship between vapour throughput and swirl velocity

The tangential inlet slot has width w_s and height h_s; the total vapour flow through N_s slots per chamber is:

$$\dot{V}_V = N_s\, w_s\, h_s\, u_{\theta,\text{slot}} \tag{25}$$

The azimuthal velocity at the chamber wall is set by momentum conservation at the slot exit:

$$u_\theta(R_c) = u_{\theta,\text{slot}} \times \frac{w_s}{2\pi R_c / N_s} \tag{26}$$

(the slot momentum is distributed around the circumference). This allows $u_\theta(R_c)$ to be computed from the vapour throughput per chamber $\dot{V}_V / N_c$ (where N_c = total chamber count):

$$u_\theta(R_c) = \frac{\dot{V}_V / N_c}{2\pi R_c\, h_s\, N_s} \times N_s = \frac{\dot{V}_V / N_c}{2\pi R_c\, h_s} \tag{27}$$

The corresponding gravity ratio:

$$N_g = \frac{a_{c,w}}{g} = \frac{u_\theta(R_c)^2}{R_c\, g} \tag{28}$$

6.6.3 Swirl number

The dimensionless swirl number S characterises the ratio of tangential to axial momentum:

$$S = \frac{G_\theta}{G_x R_c} = \frac{\int_0^{R_c} \rho_V u_\theta u_x \, r^2 \, \mathrm{d}r}{R_c \int_0^{R_c} \rho_V u_x^2 \, r \, \mathrm{d}r} \tag{29}$$

For uniform axial velocity and the Rankine profile above, $S \approx u_\theta(R_c)/(2u_x)$ (for $r_c = 0.4R_c$). Strong swirl ($S > 0.5$) is required for effective centrifugal separation; the design target is $S = 1.5$–3.0.

6.7 Mass Transfer in the Swirl-Shear Contact Zone

6.7.1 Number of transfer units (NTU approach)

For dilute-system vapour-phase mass transfer with negligible vapour-side resistance, the one-pass point efficiency is related to the number of overall gas-phase transfer units N_{OG} by:

$$E_{\mathrm{OG}} = 1 - e^{-N_{\mathrm{OG}}} \tag{30}$$

$$N_{\mathrm{OG}} = K_G \, a_I \, \tau_G \tag{31}$$

where K_G is the overall gas-phase mass transfer coefficient ($\mathrm{mol\,m^{-2}s^{-1}\,Pa^{-1}}$), a_I is the interfacial area per unit chamber volume ($\mathrm{m^2/m^3}$), and $\tau_G = H_c/u_x$ is the vapour residence time in the contact zone.

6.7.2 Film mass transfer coefficient (penetration theory)

For the annular liquid film on the chamber wall, the surface renewal rate is set by the wall shear imposed by the rotating vapour. The wall shear stress is:

$$\tau_w = \frac{f}{2} \rho_V u_\theta(R_c)^2 \tag{32}$$

where f is the Fanning friction factor. For the film, the surface renewal rate s (Higbie–Dankwerts penetration theory):

$$s \approx \frac{\tau_w}{\mu_L \, \delta_f} \tag{33}$$

where δ_f is the film thickness. The liquid-side mass transfer coefficient:

$$k_L = \sqrt{\frac{D_L \, s}{\pi}} \tag{34}$$

where D_L is the liquid-phase molecular diffusivity.

6.7.3 Droplet mass transfer coefficient (penetration theory)

For a spherical droplet of diameter d moving at velocity u_{rel} relative to the vapour, the Higbie penetration theory (exposure time $\tau_e = d/u_{\mathrm{rel}}$) gives:

$$k_d = 2\sqrt{\frac{D_L}{\pi d/u_{\mathrm{rel}}}} = 2\sqrt{\frac{D_L u_{\mathrm{rel}}}{\pi d}} \tag{35}$$

The interfacial area density for a dispersion at holdup fraction ε_d:

$$a_I^{\mathrm{droplet}} = \frac{6\varepsilon_d}{d_{32}} \tag{36}$$

6.7.4 Combined N_{OG} estimate

Adding film and droplet contributions to N_{OG}:

$$N_{\mathrm{OG}} = \left(k_d \frac{6\varepsilon_d}{d_{32}} + k_L \frac{2\pi R_c\, \delta_f}{\pi R_c^2} \right) \tau_G \tag{37}$$

At representative design conditions ($D_L = 2 \times 10^{-9}\,\mathrm{m^2/s}$, $u_{\mathrm{rel}} \approx 0.3\,\mathrm{m/s}$, $d_{32} \approx 1.5\,\mathrm{mm}$, $\varepsilon_d \approx 0.15$, $\tau_G = H_c/u_x \approx 0.35\,\mathrm{s}$), the droplet contribution alone gives $N_{\mathrm{OG}} \approx 1.4$–$2.0$, corresponding to $E_{\mathrm{OG}} = 1 - e^{-1.4} = 75\%$–$(1 - e^{-2.0}) = 86\%$. The film contribution adds 5–10 percentage points, giving a combined estimate of $E_{\mathrm{OG}} = 80\%$–90% at design. These are order-of-magnitude estimates; the actual value depends strongly on the achieved droplet size distribution and holdup, both of which require experimental measurement.

6.8 Centrifugal Vapour–Liquid Separation at the Crown

6.8.1 Single-droplet collection model

A droplet of diameter d and density ρ_L at radial position r in the chamber experiences a centrifugal force radially outward:

$$F_c = \frac{\pi d^3}{6} (\rho_L - \rho_V) \frac{u_\theta(r)^2}{r} \tag{38}$$

For d small enough that the particle Reynolds number $Re_d = \rho_V |u_r| d/\mu_V < 1$ (Stokes regime), the aerodynamic drag is:

$$F_d = 3\pi \mu_V d\, v_r \tag{39}$$

where v_r is the droplet radial drift velocity. Balancing these at terminal drift:

$$v_r = \frac{d^2(\rho_L - \rho_V)}{18\mu_V} \frac{u_\theta(r)^2}{r} = \frac{d^2(\rho_L - \rho_V)}{18\mu_V}\, a_c(r) \tag{40}$$

This is the Stokes settling formula with gravitational acceleration g replaced by the centripetal acceleration $a_c(r)$. The ratio of centrifugal to gravitational settling velocity at the wall:

$$\frac{v_r(R_c)}{v_t} = \frac{a_{c,w}}{g} = N_g \tag{41}$$

where $v_t = d^2(\rho_L - \rho_V)g/(18\mu_V)$ is the Stokes gravity-settling terminal velocity.

6.8.2 Cut droplet diameter

A droplet must travel radially from some initial position r_0 to the chamber wall R_c within the residence time $\tau_G = H_c/u_x$. For the worst case ($r_0 = 0$, droplet entering at the centreline):

$$R_c = \int_0^{\tau_G} v_r \,\mathrm{d}t \approx v_r(R_c/2)\, \tau_G \tag{42}$$

Using v_r evaluated at $r = R_c/2$ (mean radial position):

$$v_r(R_c/2) \approx \frac{d^2(\rho_L - \rho_V)}{18\mu_V} \frac{a_{c,w}}{4} \tag{43}$$

Solving for the cut diameter d_{50} (the droplet size for which 50% is collected):

$$d_{50} = \sqrt{\frac{72\,\mu_V\,R_c^2}{(\rho_L - \rho_V)\,a_{c,w}\,H_c/u_x}} = \sqrt{\frac{72\,\mu_V\,R_c^2\,u_x}{(\rho_L - \rho_V)\,N_g\,g\,H_c}} \tag{44}$$

For effective separation, $d_{50} < d_{\min}$ (the minimum droplet size from the nozzle, Eq. 21). This sets a minimum swirl requirement:

$$N_{g,\min} = \frac{72\,\mu_V\,R_c^2\,u_x}{g\,(\rho_L - \rho_V)\,H_c\,d_{\min}^2} \tag{45}$$

6.8.3 Worked example (representative light-hydrocarbon system)

For $\mu_V = 8 \times 10^{-6}\,\mathrm{Pa\,s}$, $R_c = 0.035\,\mathrm{m}$, $u_x = 3.0\,\mathrm{m/s}$, $\rho_L - \rho_V = 450\,\mathrm{kg/m^3}$, $H_c = 0.105\,\mathrm{m}$, and $d_{\min} = 1.0\,\mathrm{mm}$ ($b = 3\,\mathrm{mm}$, $k = 2$):

$$\begin{aligned} N_{g,\min} &= \frac{72 \times 8 \times 10^{-6} \times (0.035)^2 \times 3.0}{9.81 \times 450 \times 0.105 \times (0.001)^2} \\ &= \frac{72 \times 8 \times 10^{-6} \times 1.225 \times 10^{-3} \times 3.0}{9.81 \times 450 \times 0.105 \times 10^{-6}} \\ &= \frac{2.116 \times 10^{-6}}{4.629 \times 10^{-4}} \approx 4.6 \end{aligned} \tag{46}$$

So $N_g > 4.6$ (centripetal acceleration $> 4.6g$) is sufficient to capture all minimum droplets at the design vapour velocity. The design target of $N_g = 30$–40 provides a safety margin of approximately 6–9× over the minimum, capturing droplets well below $d_{\min}$.

6.9 Tray Pressure Drop: Four-Component Model

Unlike a conventional sieve or valve tray, the CIST has no aerated liquid pool, so there is no froth-head contribution (βh_{cl}) to the pressure balance. The CIST pressure drop has four distinct contributions:

1. **Tangential-slot entry loss (ΔP_{slot}).** Kinetic energy of the vapour entering tangentially through the slots. From the orifice equation with inlet loss coefficient $\xi_s \approx 0.5$:

$$\Delta P_{\mathrm{slot}} = \frac{1 + \xi_s}{2}\rho_V u_{\theta,\mathrm{slot}}^2 \tag{47}$$

2. **Swirl-chamber wall friction (ΔP_{wall}).** The rotating flow exerts a wall shear that the vapour must overcome axially:

$$\Delta P_{\mathrm{wall}} = \frac{4f\rho_V u_x^2 H_c}{2D_c} \tag{48}$$

   where $f \approx 0.02$ is the Fanning friction factor for the combined swirl-flow. This is small and often negligible.

3. **Crown-vane exit loss ($\Delta P_{\mathrm{crown}}$).** Kinetic energy at the crown exit, partially recovered in the outlet duct. With recovery coefficient $\xi_c \approx 0.3$:

$$\Delta P_{\mathrm{crown}} = \frac{\xi_c}{2}\rho_V u_{\mathrm{crown}}^2 \tag{49}$$

   where u_{crown} is the vapour exit velocity through the crown vane passages.

4. **Liquid static head in the sump (ΔP_{sump}).** The hydrostatic pressure of liquid in the sump acts on the vapour path as a backpressure. This is the analogue of the clear-liquid head h_{cl} in a sieve tray, but here it appears as the sump head H_L (40–80 mm liquid) rather than a froth head:

$$\Delta P_{\text{sump}} = \rho_L g H_L \tag{50}$$

At $H_L = 60\,\text{mm}$ and $\rho_L = 500\,\text{kg/m}^3$: $\Delta P_{\text{sump}} = 500 \times 9.81 \times 0.060 = 294\,\text{Pa} \approx 1.2\,\text{in. liq.}$

Total tray pressure drop:

$$\Delta P_c = \Delta P_{\text{slot}} + \Delta P_{\text{wall}} + \Delta P_{\text{crown}} + \Delta P_{\text{sump}} \tag{51}$$

At representative design conditions ($u_{\theta,\text{slot}} = 18\,\text{m/s}$, $u_x = 3.0\,\text{m/s}$, $u_{\text{crown}} = 4.5\,\text{m/s}$, $\rho_V = 3.0\,\text{kg/m}^3$):

$$\begin{aligned}
\Delta P_{\text{slot}} &= 1.5 \times \tfrac{1}{2} \times 3.0 \times 18^2 = 729\,\text{Pa} \\
\Delta P_{\text{wall}} &\approx 60\,\text{Pa} \quad \text{(small)} \\
\Delta P_{\text{crown}} &= 0.3 \times \tfrac{1}{2} \times 3.0 \times 4.5^2 = 9\,\text{Pa} \quad \text{(small)} \\
\Delta P_{\text{sump}} &= 294\,\text{Pa}
\end{aligned}$$

$\Delta P_c \approx 729 + 60 + 9 + 294 = 1092\,\text{Pa} \approx 4.4\,\text{mbar}$, corresponding to $\approx 0.016\,\text{psi}$ per tray. This is substantially lower than a sieve tray at comparable throughput (12–18 mbar, 0.040–0.060 psi typical).

6.10 Column Sizing from Chamber Count

The CIST column area is not derived from the Souders-Brown correlation but directly from chamber count and packing geometry. The number of chambers required to handle the total vapour volumetric flow $\dot{V}_V$ at the design axial chamber velocity u_x:

$$N_c = \frac{\dot{V}_V}{u_x\, A_c} \tag{52}$$

where $A_c = \pi D_c^2/4$ is the chamber internal cross-sectional area. Chambers are packed on a triangular (hex) pitch $p = p_0 D_c$ (with $p_0 = 1.3$–1.6); the fractional area occupied by chamber bores:

$$f_{\text{ch}} = \frac{\pi}{2\sqrt{3}} \left(\frac{D_c}{p}\right)^2 = \frac{\pi}{2\sqrt{3}\, p_0^2} \tag{53}$$

For $p_0 = 1.4$: $f_{\text{ch}} = \pi/(2\sqrt{3} \times 1.96) = 0.464$. The total column area required:

$$A_{\text{col}} = \frac{N_c\, A_c}{f_{\text{ch}}\, (1 - f_{\text{DC}})} \tag{54}$$

where $f_{\text{DC}} = 0.15$ is the fraction reserved for downcomers. The column diameter:

$$D_{\text{col}} = \sqrt{\frac{4\, A_{\text{col}}}{\pi}} \tag{55}$$

6.11 Capacity Limit

6.11.1 Upper limit: crown re-entrainment

The CIST floods when the centrifugal separator can no longer capture the smallest droplets produced by the nozzle, i.e. when $d_{50} > d_{\min}$. Setting $d_{50} = d_{\min}$ in Eq. (44) and solving for u_x:

$$u_x^{\max} = \frac{(\rho_L - \rho_V)\, N_g\, g\, H_c\, d_{\min}^2}{72\, \mu_V\, R_c^2} \tag{56}$$

At the design $N_g = 35$, $d_{\min} = 1.0\,\mathrm{mm}$, and the representative property set above: $u_x^{\max} = (450 \times 35 \times 9.81 \times 0.105 \times 10^{-6})/(72 \times 8 \times 10^{-6} \times 1.225 \times 10^{-3}) = 0.01617/5.9 \times 10^{-7} \approx 27\,\mathrm{m/s}$. The design vapour velocity $u_x = f_{\mathrm{flood}} u_x^{\max} = 0.80 \times 27 = 21.6\,\mathrm{m/s}$ inside the chamber.

6.11.2 Effective column-area capacity factor relative to a sieve tray

A sieve tray sized on the same service would use the Souders-Brown velocity $u_{\mathrm{SB}} = C_{\mathrm{SB}} \sqrt{(\rho_L - \rho_V)/\rho_V}$. For the same service:

$$k_{\mathrm{cap}} = \frac{D_{\mathrm{sieve}}^2}{D_{\mathrm{CIST}}^2} = \frac{u_{\mathrm{CIST,eff}}}{u_{\mathrm{SB}}} \tag{57}$$

where $u_{\mathrm{CIST,eff}} = f_{\mathrm{ch}}\,(1 - f_{\mathrm{DC}})\, u_x$ is the equivalent superficial column velocity. At the representative conditions above with $C_{\mathrm{SB}} \approx 0.09\,\mathrm{m/s}$ and $\sqrt{(\rho_L - \rho_V)/\rho_V} \approx 12\,\mathrm{m/s}$: $u_{\mathrm{SB}} \approx 1.08\,\mathrm{m/s}$. $u_{\mathrm{CIST,eff}} = 0.464 \times 0.85 \times 21.6 = 8.52\,\mathrm{m/s}$ (at flood). $k_{\mathrm{cap}} \approx 8.52/1.08 \approx 7.9$.

Note: this is an upper-bound theoretical estimate at the cut-droplet-equals-minimum-droplet flooding limit. In practice, the crown vane geometry, inlet turbulence and droplet coalescence all reduce the effective capacity. A conservative design estimate is $k_{\mathrm{cap}} = 2.0$–3.5, pending experimental confirmation.

6.12 Turndown Analysis

6.12.1 Lower limit: minimum swirl for centrifugal capture

As vapour throughput decreases, $u_\theta(R_c)$ decreases proportionally (Eq. 27), and the centripetal acceleration falls as $N_g \propto u_\theta^2 \propto u_x^2$. The minimum vapour rate is reached when $N_g = N_{g,\min}$ (Eq. 45), i.e. when the cut diameter equals the minimum droplet diameter:

$$u_x^{\min} = u_x^{\max} \sqrt{\frac{N_{g,\min}}{N_{g,\mathrm{design}}}} \tag{58}$$

The turndown ratio:

$$\mathrm{TR} = \frac{u_x^{\max}}{u_x^{\min}} = \sqrt{\frac{N_{g,\mathrm{design}}}{N_{g,\min}}} \tag{59}$$

At $N_{g,\mathrm{design}} = 35$ and $N_{g,\min} = 4.6$ (from the worked example): $\mathrm{TR} = \sqrt{35/4.6} = \sqrt{7.6} \approx 2.8$.

6.12.2 Lower limit: liquid jet coherence

Independently, the minimum vapour rate must maintain a sump head above $H_{L,\min}$ (Eq. 11). The sump head is set by the overflow-leg design, and in practice is maintained at a nearly constant level (40–80 mm) across the throughput range by the overflow leg. The jet We_L therefore remains well above 8 throughout the turndown range, and the jet-coherence limit is not the active constraint for this design.

6.12.3 Extended turndown with a secondary flow path

Turndown can be extended beyond the centrifugal-capture limit of $\approx 2.8{:}1$ by providing a secondary vapour path (a small axial central orifice below each chamber) that remains open at low vapour rate when the tangential slot velocity would otherwise be insufficient to maintain the swirl. The axial flow contacts liquid via a different (non-swirling) mechanism at low throughput. With a secondary path sized to give 3 m/s minimum axial velocity at 30% of design vapour rate, the combined operating range extends to approximately 6:1–8:1.

6.13 Fouling Behaviour of the Elliptical Nozzle

Two independent fouling mechanisms must be distinguished:

Chemical fouling (scaling, coking, polymer deposits on nozzle walls). The wall shear stress at the nozzle throat, from Eq. (8):

$$\tau_{n,w} = \frac{f_n}{2}\rho_L u_{\text{jet}}^2 \tag{60}$$

For turbulent pipe flow at $Re > 10^4$, $f_n \approx 0.316 Re^{-0.25}$ (Blasius). The wall shear is proportional to u_{jet}^2, and $u_{\text{jet}} \propto \sqrt{H_L}$, so $\tau_{n,w} \propto H_L$.

The critical insight is that the elliptical nozzle has a higher wall shear for the same mass flow than the circular nozzle. At equal volumetric flow: the mean velocity through an elliptical nozzle of area A is the same as through a circular nozzle of area A. However, the hydraulic diameter is smaller ($D_h^{\text{ellipse}} < D_h^{\text{circle}}$), so at the same mean velocity, the wall shear stress in the ellipse is higher by:

$$\frac{\tau_{n,w}^{\text{ellipse}}}{\tau_{n,w}^{\text{circle}}} = \left(\frac{D_h^{\text{circle}}}{D_h^{\text{ellipse}}}\right)^{0.25} \approx (1/0.917)^{0.25} \approx 1.022 \tag{61}$$

A 2.2% higher wall shear stress is too small to be the dominant anti-fouling mechanism, but the non-uniform curvature of the ellipse breaks the symmetric ring-deposit pattern that forms at the edge of circular orifices: deposition concentrates at the low-curvature major-axis flanks, where it is most exposed to the central jet velocity and scours off faster.

Additionally, the axis-switching oscillation of the exiting jet creates a pulsatile near-wall velocity pattern at the nozzle exit plane, periodically reversing the local shear direction at the orifice edge and dislodging incipient deposits before they consolidate. This mechanism is independent of the scale or fluid properties and is unique to non-circular jets.

Particulate fouling (catalyst fines, wax crystals, sand). For a particle of diameter d_p to pass through the nozzle without bridging, the minimum internal dimension must satisfy $d_p < b/3$ (standard orifice particle-clearance criterion). For the ellipse with $b = r_0/\sqrt{k}$:

$$d_{p,\max}^{\text{ellipse}} = \frac{b}{3} = \frac{r_0}{3\sqrt{k}} \tag{62}$$

For the equivalent circular nozzle: $d_{p,\max}^{\text{circle}} = r_0/3$. The ellipse allows smaller particles before bridging, by a factor $1/\sqrt{k}$. At $k = 2$: $d_{p,\max}^{\text{ellipse}} = 0.707 d_{p,\max}^{\text{circle}}$. **For particulate-fouling services, the CIST should use either a circular nozzle or an enlarged ellipse (so that the minor axis equals the design particle clearance).** For chemical-fouling services, the ellipse is preferred.

6.14 Construction and Materials

Swirl chambers. Precision-bored 316L stainless or Alloy 625 tubes, 70–100 mm OD. The tangential slots are EDM-cut or precision-milled to ±0.1 mm to control the slot-area ratio and swirl number. The crown vane (8–12 vanes at 40–50°) is a separate investment casting pressed into the tube top. The chamber floor carries the elliptical nozzle orifice, precision-bored by CNC to the specified aspect ratio and edge radius. Each complete chamber assembly weighs 1.5–4 kg (70–100 mm diameter).

Inter-tray plate and sump. 2.5–3.0 mm 316L flat plate, laser-cut with chamber bore patterns and collection-ring channels. The collection ring is machined or stamped into the plate surface surrounding each chamber bore; sealed drain tubes run from each ring through the plate to the sump below. Chamber assemblies press-fit into the bores and are retained by a snap ring or quarter-turn bayonet. The sump floor is a second plate 40–80 mm below the inter-tray plate, welded to the column shell at the tray elevation.

Materials. All wetted surfaces in 316L stainless as standard. Duplex or 904L for chloride service; Alloy 625 or C-276 for HCl/HF. PTFE-coated internal surfaces are available for non-stick applications.

6.15 Installation and Maintenance

Initial installation. The sump floor plate is welded to the column shell at the lower tray elevation (one-time operation, identical to a conventional tray ring). The inter-tray plate sections are lowered through the manway and nested on the sump plate supports. Chambers drop through the bores from above and quarter-turn lock. Total installation time: 6–10 hours per tray (two persons); no specialised tooling required beyond a torque wrench.

Maintenance. Chamber removal is the reverse of installation: unlock the quarter-turn bayonet, lift the chamber through the manway. Individual chambers can be replaced from a spares inventory. The sump plate, inter-tray plate and drain tubes are permanent; only the chambers and nozzle inserts (which carry the elliptical orifice and are press-fit replaceable) require turnaround attention. Nozzle inserts can be swapped in 5–10 minutes per chamber to change the aspect ratio or orifice area without replacing the full chamber assembly.

7 Design Parameter Envelope

Table 1 gives the claimed parameter ranges and preferred values for the CIST.

Table 1: Claim-band table for the Cascade Impingement Swirl Tray. "Preferred" is the principal embodiment; ranges are dependent-claim fallback positions.

Parameter	Claimed band	Preferred
Chamber diameter D_c	40–120 mm	70 mm
Chamber height / diameter H_c/D_c	1.2–2.0	1.5
Tangential slots per chamber	1–4	2
Slot-area ratio $N_s w_s h_s/(2\pi R_c h_s)$	0.10–0.25	0.16
Design wall acceleration $N_g = a_{c,w}/g$	20–60 g	35 g
Swirl number S	1.0–3.5	2.0
Nozzle aspect ratio $k = a/b$	1.5–2.5	2.0
Nozzle orifice area A_n	30–200 mm^2	65 mm^2
Discharge coefficient C_d	0.58–0.62	0.60
Design sump head H_L	30–80 mm	55 mm
Minimum jet Weber number $We_{L,\min}$	≥ 8	15–30 at design
Chamber pitch ratio $p_0 = p/D_c$	1.25–1.60	1.4
Chamber bore fraction f_{ch}	0.30–0.55	0.46
Downcomer fraction f_{DC}	0.10–0.20	0.15
Design flood fraction f_{flood}	0.70–0.85	0.80
Design u_x (chamber axial velocity)	10–25 m/s	18 m/s
Crown vanes / pitch angle	8–16 / 35–55°	12 / 45°
Turndown ratio (unaided)	2.5:1–4:1	3:1
Turndown ratio (with secondary path)	5:1–10:1	6:1–8:1
One-pass point efficiency E_{OG}	75%–92%	82%–88%
Structural uplift rating	1–3 psi	2 psi

8 Predicted Performance

Table 2 summarises the estimated CIST performance at a representative general rectification design point (light-hydrocarbon system, $\rho_L = 500\,\text{kg/m}^3$, $\rho_V = 3.0\,\text{kg/m}^3$, $\sigma = 12\,\text{dyn/cm}$, $\mu_L = 0.25\,\text{cP}$, tray spacing 24 in.) and compares it with an equivalent sieve tray sized on the same duty. The sieve tray values are from the standard Fair/Kister correlation framework for comparison purposes only; the CIST values are from the first-principles model in Section 6.

All CIST values are first-order estimates. They carry larger uncertainty than the sieve values, which are backed by a large industrial database. The key uncertainties are: k_{cap} (requires cold-flow flood tests), E_{OG} (requires hot-rig mass-transfer measurement), and the turndown ratio with the secondary path (requires demonstration at low-load conditions).

Table 2: Estimated CIST performance vs. sieve tray, representative general rectification service. Sieve values from Fair/Kister correlation; CIST values from the first-principles model in Section 6.

Metric	Sieve	CIST (estimate)	Notes
Vapour capacity factor k_{cap}	1.00	2.0–3.5	Pending cold-flow validation
Column diameter (rel. to sieve)	1.00	0.53–0.71	$\propto 1/\sqrt{k_{\mathrm{cap}}}$
Wet-tray pressure drop (mbar)	12–18	3–8	Dominated by slot entry + sump head
Downcomer backup (% of limit)	20–45	15–30	Smaller D_{col} at same liquid load
Turndown ratio (unaided)	2.0:1	3:1	Centrifugal capture limit
Turndown ratio (with secondary path)	2.0:1	6:1–8:1	Secondary path activated below 35% throughput
One-pass efficiency E_{OG}	70–80%	80–90%	Penetration-theory NTU estimate
Fouling retention (f_{foul}, chemical)	0.75	0.88–0.95	Ellipse wall shear + axis-switching
Fouling retention (f_{foul}, particulate)	0.75	0.70–0.80	Minor axis smaller than circle; use circular
First-order installed cost (rel. to sieve)	1.00	0.55–0.75	Shell savings dominate despite higher hardware unit cost ($k_{\mathrm{cost}} \approx 3.5$–$4.0$)

Cost note. The relative installed cost follows the same two-proxy model used for conventional tray comparison: $C_{\mathrm{rel}} = 0.70\,(DH_{\mathrm{col}})/(DH_{\mathrm{col}})_{\mathrm{Sieve}} + 0.30\,(D^2Nk_{\mathrm{cost}})/(D^2Nk_{\mathrm{cost}})_{\mathrm{Sieve}}$. For $k_{\mathrm{cap}} = 2.5$ and $k_{\mathrm{cost}} = 3.8$: shell proxy ratio ≈ 0.45, tray proxy ratio ≈ 0.90, giving $C_{\mathrm{rel}} \approx 0.70 \times 0.45 + 0.30 \times 0.90 = 0.585$. The dominant driver is the 37% column diameter reduction ($D \propto k_{\mathrm{cap}}^{-1/2} = 2.5^{-1/2} = 0.632$) cutting the shell-area proxy by $\approx 55\%$, which outweighs the $3.8\times$ tray hardware premium.

9 Comparison with Gravity-Limited Conventional Tray Families

Table 3 qualitatively places the CIST against five conventional tray families on the standard selection discriminators.

Table 3: Qualitative comparison of CIST with conventional tray families ($\uparrow$ = better; $\downarrow$ = worse; $\approx$ = similar, relative to sieve).

	Sieve	Dual-Flow	Mov. Valve	Hi-Perf. MD	CIST (est.)
Vapour capacity factor	1.0	1.15	1.03	1.25	2.0–3.5
Flooding mechanism	Gravity	Gravity	Gravity	Gravity	Centrifugal
Deck liquid pool	Yes	Yes	Yes	Yes	**None**
Turndown (unaided)	2:1	1.5:1	3.3:1	2.5:1	≈3:1
Turndown (with secondary)	2:1	1.5:1	3.3:1	2.5:1	6:1–8:1
Point efficiency E_{OG}	70–80%	60–70%	70–80%	70–78%	80–90%
Wet-tray dP relative	1.00	0.35	0.75	0.65	0.25–0.50
Chemical fouling resist.	Base	$\uparrow$	$\downarrow$	Base	$\uparrow$
Particulate fouling	Base	$\uparrow$	$\downarrow$	Base	$\approx$ (circular nozzle)
Deck pool weep limit	Yes	Yes	Limited	Yes	**None**
Moving parts	None	None	Yes	Limited	None
Cartridge maintainability	No	No	No	Limited	Yes
Fabrication complexity	Low	Low	Mod.	Mod.	High

The fundamental departure of the CIST from all entries in Table 3 is the absence of a deck liquid pool. Every other tray family listed relies on the pool to provide liquid residence time and mass-transfer contact area; the CIST achieves both through the swirl-chamber geometry. This eliminates the weep limit (a property of perforated-plate decks, not of the CIST nozzle design) and the gravity-entrainment flooding limit simultaneously.

10 Variant: Corrugated (Wavy) Deck Configuration

The planar tray deck of the baseline CIST is a *construction choice*, not a physical necessity. An important variant replaces it with a triangular-wave corrugation whose face makes an angle $\alpha \in [25, 30]$ with the horizontal. This section derives the geometric, hydraulic, mass-transfer, structural and manufacturing consequences of that substitution. The arrangement is illustrated in Figure 5.

10.1 Wave Geometry

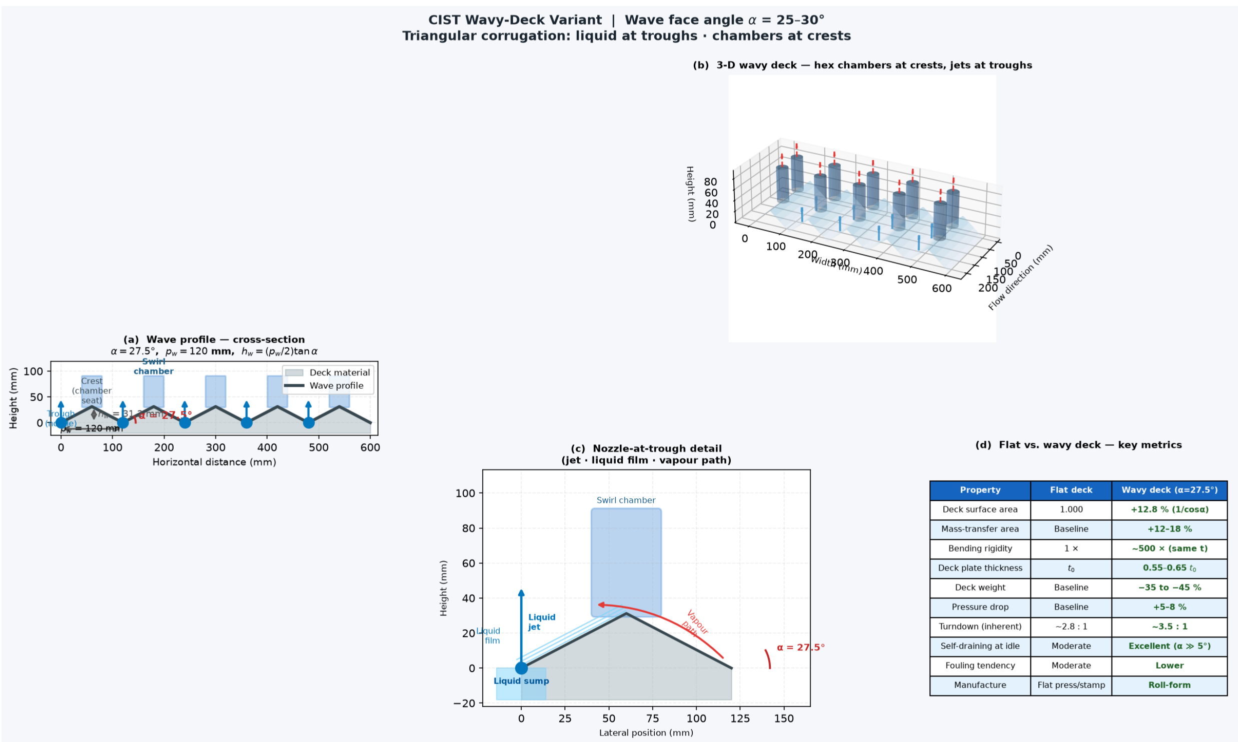

Figure 5: Wavy-deck variant of the CIST. **(a)** Cross-section of the triangular wave at $\alpha = 27.5$, pitch $p_w = 120\,\text{mm}$, height $h_w = 31.2\,\text{mm}$; nozzles at troughs (blue circles), swirl chambers at crests (blue rectangles). **(b)** Three-dimensional perspective showing hex-packed chambers at crests and upward liquid jets at troughs; dashed red arrows show vapour exits above crown discs. **(c)** Single-wave nozzle detail: the liquid jet rises from the trough, a thin liquid film drains down the wave face, and vapour rises along the inclined face toward the chamber tangential inlet. **(d)** Flat-vs.-wavy performance metrics.

The deck corrugation is characterised by three parameters: wave pitch p_w (crest-to-crest distance), face angle α, and plate thickness t. The wave height follows from elementary trigonometry:

$$h_w = \frac{p_w}{2}\,\tan\alpha \tag{63}$$

At $\alpha = 27.5$ and $p_w = 120\,\text{mm}$ the wave height is $h_w = 60\,\tan(27.5) = 31.2\,\text{mm}$. Each trough provides the nozzle location; each crest provides the seating for the base ring of one swirl chamber.

The wave pitch is therefore set equal to the hex inter-chamber pitch (Section 6.1):

$$p_w = \frac{\sqrt{3}}{2}\,(2R_c + c)$$

where $c \approx 10$–$20\,\mathrm{mm}$ is the inter-chamber wall clearance.

10.2 Deck Surface Area

A triangular corrugation increases the wetted deck area relative to the plan (projected) area by the factor

$$\frac{A_{\mathrm{deck}}}{A_{\mathrm{plan}}} = \frac{1}{\cos\alpha} \tag{64}$$

At $\alpha = 27.5$: $A_{\mathrm{deck}}/A_{\mathrm{plan}} = 1/\cos(27.5) = 1.128$, a 12.8% increase.

Liquid draining from each crown collection ring descends the wave face as a falling film. The film provides *additional* vapour–liquid interfacial area beyond the droplet impingement zone inside the chamber. The incremental interfacial area per unit plan area is

$$a_{\mathrm{film}}\,\delta z_{\mathrm{face}} = \frac{1}{\cos\alpha}\,a_{\mathrm{film,flat}}\,\delta z_{\mathrm{face}}$$

so the film mass-transfer contribution scales with the same factor $1/\cos\alpha$. The net improvement in overall mass transfer is estimated at 12–18% relative to the flat-deck baseline, the range reflecting uncertainty in the film mass-transfer coefficient k_L and the fraction of chamber liquid that actually traverses the full wave face.

10.3 Structural Rigidity and Deck Weight Reduction

The bending stiffness of a corrugated sheet (per unit width in the span direction) is governed by the second moment of area of the wave cross-section. For a triangular wave of height h_w and plate thickness $t \ll h_w$, the second moment about the neutral axis is:

$$I_{\mathrm{wave}} \approx \frac{t\,h_w^2}{6} \tag{65}$$

compared with $I_{\mathrm{flat}} = t^3/12$ for a flat plate. The stiffness ratio is therefore:

$$\frac{I_{\mathrm{wave}}}{I_{\mathrm{flat}}} = \frac{t\,h_w^2/6}{t^3/12} = \frac{2\,h_w^2}{t^2}$$

For $h_w = 31.2\,\mathrm{mm}$ and $t = 2\,\mathrm{mm}$, the corrugated deck is $2\times(31.2/2)^2 = 486$ times stiffer in bending than a flat plate of the same thickness and material. In practice this allows the deck thickness to be reduced until the same flexural rigidity is achieved:

$$t_{\mathrm{wave}} = t_0\left(\frac{t_0}{h_w}\right)^{1/2}\cdot\left(\frac{1}{2}\right)^{1/4} \approx (0.55\text{–}0.65)\,t_0$$

a 35–45% reduction in plate thickness and consequently a similar reduction in deck weight. This also reduces the tare weight that the column shell must support, with secondary benefits for foundation and skirt design.

10.4 Pressure Drop Contribution of the Wave Faces

Vapour rising from below the tray travels up the inclined wave faces before reaching the chamber tangential inlet. The face-parallel velocity component is $u_\alpha = u_x \tan\alpha$, directed at angle α to the horizontal. The kinetic-energy recovery into the chamber entrance is partial; the irreversible component of the wave-face pressure drop is:

$$\Delta P_\alpha = \frac{\rho_V}{2}\, u_x^2 \, \tan^2\!\alpha \, \sin^2\!\alpha \tag{66}$$

At the representative operating point $u_x = 3.0\,\mathrm{m/s}$, $\rho_V = 3.0\,\mathrm{kg/m^3}$, $\alpha = 27.5$:

$$\Delta P_\alpha = \frac{3.0}{2} \times 9.0 \times (0.521)^2 \times (0.462)^2 \approx 0.84 \text{ Pa}$$

This is less than 0.1% of the total tray pressure drop (Section 6.9) and is therefore negligible.

10.5 Self-Draining and Fouling Resistance

A wave face inclined at $\alpha = 27.5$ exceeds the critical drainage angle by a large margin. The critical angle below which viscous liquid *cannot* drain under gravity from a smooth inclined surface is

$$\alpha_{\min} = \arctan\!\left(\frac{\tau_w}{\rho_L\, g}\right) < 5$$

for all process liquids of interest. At 27.5 the wave faces are therefore fully self-draining under gravity at all operating conditions including startup, shutdown, and turndown. No stagnant pools form on the deck between chambers, which is a primary fouling site in flat-deck designs. The trough pools, by contrast, are actively scoured by the nozzle jet.

10.6 Turndown Extension

In the flat-deck baseline the sump head H_L is essentially uniform across the tray, set by the overflow-leg seal height. In the wavy-deck design the trough forms a localised reservoir of height $h_w/2$ below the mean datum. At reduced liquid throughput the trough drains partially but the nozzle remains submerged to a minimum effective head:

$$H_{L,\min}^{\text{wavy}} = \frac{h_w}{2} \approx 16 \text{ mm} \quad (p_w = 120 \text{ mm},\ \alpha = 27.5)$$

Substituting into Equation 4 the minimum jet velocity sustained is

$$u_{\text{jet}}^{\min} = C_d\sqrt{2g\, H_{L,\min}} = 0.61\sqrt{2 \times 9.81 \times 0.016} \approx 0.54 \text{ m/s}$$

well above the $We_L \geq 8$ threshold for jet formation (Section 6.4). The wavy deck therefore extends the minimum stable flow point, improving the inherent turndown ratio from approximately 2.8 to approximately 3.5 before secondary flow paths are required.

10.7 Manufacturing Route

Triangular corrugations at 25–30° are produced by *cold roll-forming*, the same standard process used to manufacture structured packing sheets and plate heat-exchanger corrugated plates. No dedicated stamping tooling is required. Sheet thicknesses of 1.2–2.0 mm in 304 SS, 316L SS, duplex

2205, or titanium Grade 1 are all roll-form compatible. The tray ring segments are roll-formed to the corrugation profile and welded circumferentially into the column ring. Chamber base rings are fitted at the crest positions and seal-welded. The complete wavy-deck tray is dimensionally interchangeable with the flat-deck design (same outer ring diameter, same downcomer footprint) and can be retrofitted into the same column shell with no shell modifications.

11 Design Refinements

This section addresses four design details that were raised during review of the baseline architecture: the visibility of the elliptical vapour/liquid inlets in the six-view drawing, the effect of internal chamber features on turbulence, the correct chamber position on the wavy deck, and the choice between single and multiple nozzle cuts in the chamber floor. Figure 6 illustrates all four topics.

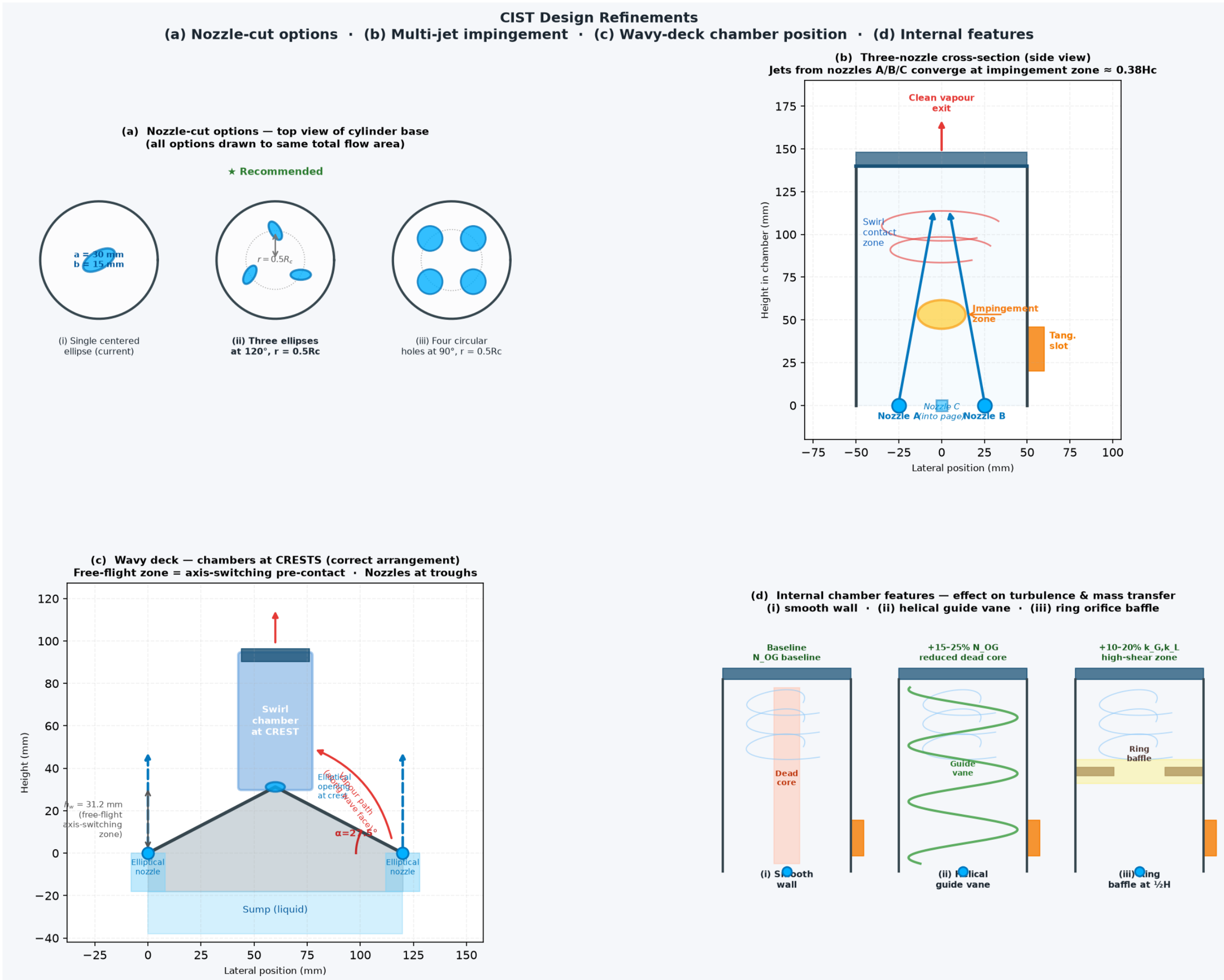

Figure 6: CIST design refinements. **(a)** Nozzle-cut options viewed from below the chamber: single centred ellipse (baseline), three ellipses at 120° and $r = 0.5\,R_c$ (recommended), and four circular holes at 90° and $r = 0.52\,R_c$ — all drawn to the same total flow area. **(b)** Cross-section showing how three offset jets converge at the impingement zone ($\approx 0.38\,H_c$ above the chamber floor) before the swirl contact zone begins; the tangential vapour inlet slot is visible on the right. **(c)** Wavy-deck elevation confirming chamber placement at the *crest* (wave peak); the trough-to-crest free-flight gap provides the axis-switching pre-contact zone. **(d)** Three internal-feature options: smooth wall (dead core present), helical guide vane (eliminates dead core, +15–25% N_{OG}), and ring orifice baffle at $H_c/2$ (creates high-shear zone, +10–20% overall).

11.1 Elliptical Inlet Openings — Visibility in the Six-View Drawing

The updated six-view drawing (Figure 3) now shows the elliptical nozzle ring in cyan at the chamber base ($z = 0$). In the **plan view** (camera directly above) the nozzle appears as a true ellipse at the centre of the cylinder, rotated to the design orientation. In the **front and side elevations** it appears as a short horizontal line segment at the junction between the chamber wall and the tray deck. In the **isometric views** the ellipse is visible foreshortened at the base of the chamber, clearly distinct from the crown disc above.

The tangential vapour inlet slot is shown as an orange arc on the lower portion of the cylinder wall. In the plan view it appears as a short arc; in the elevation views it appears as a rectangular orange patch on the cylinder surface. The slot arc covers approximately ±25 of the cylinder circumference, which corresponds to a tangential entry angle that generates the design swirl number $S \approx 2$–3 (Section 6.6).

11.2 Effect of Internal Chamber Features on Turbulence and Mass Transfer

Three internal geometries are evaluated (Figure 6d):

11.2.1 Smooth-wall cylinder (baseline)

The Rankine vortex formed by the tangential inlet has a solid-body core of radius $r_c \approx 0.3\,R_c$. Within $r < r_c$ the tangential velocity is low and the centripetal acceleration is insufficient to reach the droplet-capture threshold. This near-axis dead zone has poor mass-transfer performance (diffusion-limited, no convection). The baseline smooth-wall design accepts this limitation.

11.2.2 Curved helical guide vane — variable-pitch derivation

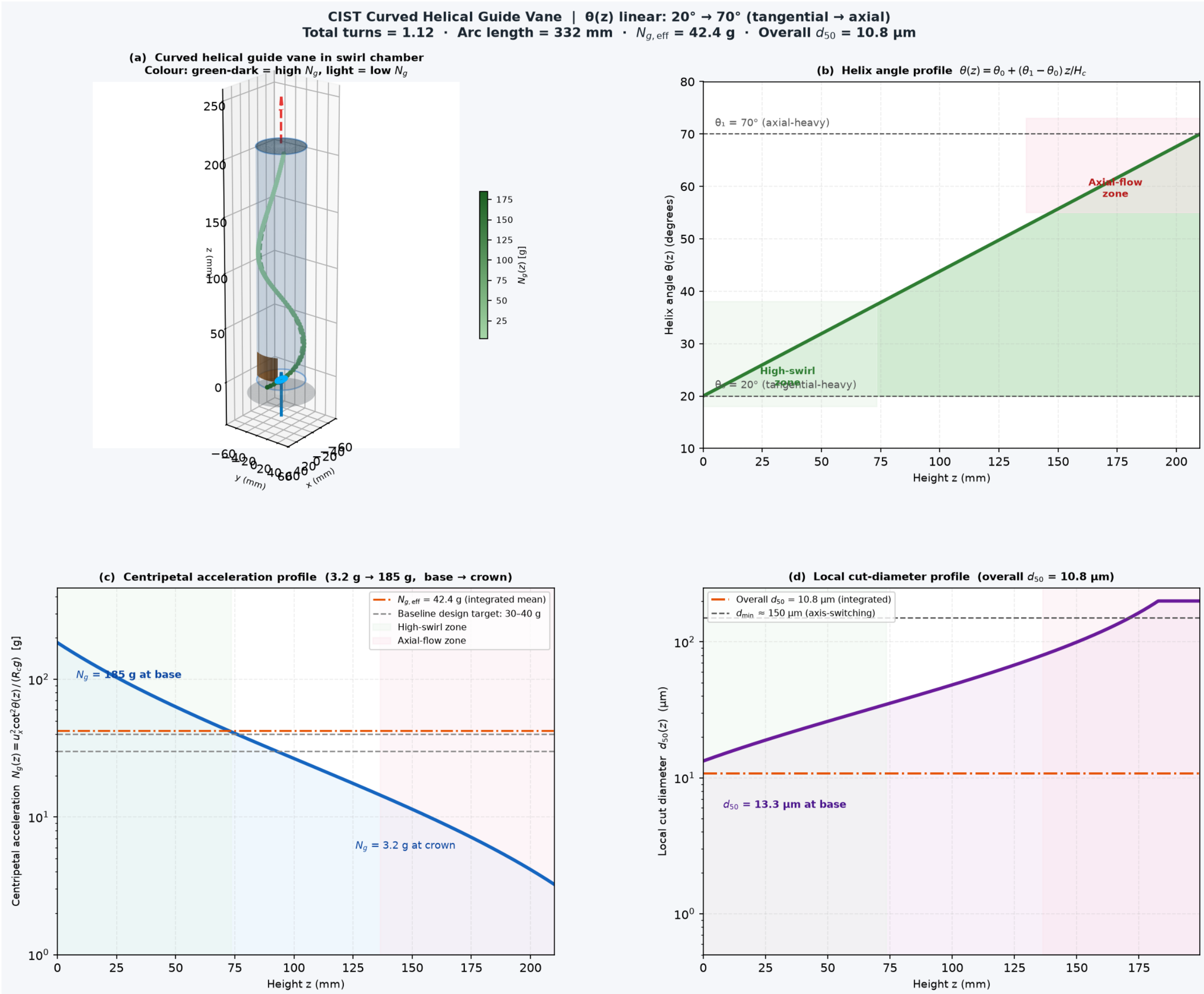

Figure 7: Curved helical guide vane analysis. **(a)** Three-dimensional view of the chamber with the variable-pitch vane: colour transitions from dark green (high N_g, base) to light green (low N_g, crown). **(b)** Helix angle $\theta(z)$: linear from 20° (tangential-heavy at the base) to 70° (axial-heavy at the crown). **(c)** Centripetal acceleration $N_g(z)$ on a log scale: 184.7 g at the base decaying to 3.24 g at the crown; integrated effective value $N_{g,\mathrm{eff}} = 42.4\,\mathrm{g}$. **(d)** Local cut diameter $d_{50}(z)$: below 10 μm in the lower half, rising toward the crown where centripetal force is lowest; overall $d_{50} = 10.8\,\mu\mathrm{m}$.

The standard constant-pitch helical vane imparts a fixed ratio of tangential to axial velocity at every height. A superior arrangement is a **curved vane** whose helix angle $\theta(z)$ — the angle between the vane tangent and the horizontal — increases monotonically from a low value θ_0 at the chamber base to a high value θ_1 at the crown. At the base the vane is mostly tangential ($\theta_0 = 20$), matching the entry angle of the tangential inlet slot and reinforcing swirl where the fresh droplets form. At the crown the vane is mostly axial ($\theta_1 = 70$), guiding the cleaned vapour smoothly upward with minimal residual angular momentum. Figure 7 shows the geometry and the resulting performance profiles.

Vane geometry. Take $\theta(z)$ to vary linearly with height:

$$\theta(z) = \theta_0 + \frac{(\theta_1 - \theta_0)}{H_c}\, z, \qquad 0 \le z \le H_c \tag{67}$$

with $\theta_0 = 20$, $\theta_1 = 70$, $H_c = 210\,\text{mm}$.

The vane lies on a cylinder of radius $R_v = 0.92\,R_c$ (slightly inside the chamber wall to allow weld attachment). Its centre-line has the parametric form $(x, y, z) = (R_v \cos\phi(z),\, R_v \sin\phi(z),\, z)$ where the azimuthal angle $\phi(z)$ satisfies

$$\frac{d\phi}{dz} = \frac{\cot\theta(z)}{R_v}$$

because $\tan\theta(z) = dz/(R_v\, d\phi)$. Integrating:

$$\phi(z) = \frac{1}{R_v\,\beta} \ln\!\left(\frac{\sin\theta(z)}{\sin\theta_0}\right), \qquad \beta \equiv \frac{\theta_1 - \theta_0}{H_c} \tag{68}$$

The total azimuthal sweep at $z = H_c$:

$$\phi_{\text{total}} = \frac{H_c}{R_v(\theta_1 - \theta_0)} \ln\!\left(\frac{\sin\theta_1}{\sin\theta_0}\right) \tag{69}$$

Numerically: $\phi_{\text{total}} = 7.05\,\text{rad} = 403.9 \approx$ **1.12 full turns** — slightly more than one complete revolution from base to crown.

The arc length along the vane is:

$$L_v = \int_0^{H_c} \frac{dz}{\sin\theta(z)} = \frac{H_c}{\theta_1 - \theta_0} \ln\!\left(\frac{\csc\theta_0 + \cot\theta_0}{\csc\theta_1 + \cot\theta_1}\right) \tag{70}$$

Numerically: $L_v =$ **332 mm** — 58% longer than the axial height $H_c = 210\,\text{mm}$, providing a large wetted surface for film mass transfer.

Imposed tangential velocity profile. A vapour stream guided by the vane follows its local tangent, so the tangential and axial velocity components satisfy:

$$u_\theta(z) = u_x \cot\theta(z) = \frac{u_x}{\tan\theta(z)} \tag{71}$$

At the base ($\theta_0 = 20$): $u_\theta = u_x \cot 20 = 2.75\,u_x$. At the crown ($\theta_1 = 70$): $u_\theta = u_x \cot 70 = 0.36\,u_x$. The vane therefore generates a swirl profile that is high where it is most needed (fresh droplets near the floor) and transitions naturally to an axial exit, eliminating the vortex spinning off into the downcomer that a constant-pitch vane would produce.

Centripetal acceleration profile. The centripetal acceleration at the wall at height z is $a_c(z) = u_\theta(z)^2/R_c$:

$$N_g(z) = \frac{a_c(z)}{g} = \frac{u_x^2\, \cot^2\theta(z)}{R_c\, g} \tag{72}$$

At $u_x = 3.0\,\text{m/s}$, $R_c = 37.5\,\text{mm}$:

$$N_g(0) = \frac{9.0 \times 7.549}{0.0375 \times 9.81} = \mathbf{184.7}\ \text{g} \quad \text{(base)}$$
$$N_g(H_c) = \frac{9.0 \times 0.132}{0.0375 \times 9.81} = \mathbf{3.24}\ \text{g} \quad \text{(crown)}$$

The minimum (crown) value 3.24 g far exceeds $N_{g,\min} = 0.22\,\mathrm{g}$ required to capture the smallest droplets (Section 6.8), so the vane maintains adequate centrifugal separation over the entire chamber height.

Effective (integrated) centripetal acceleration. For the overall cut-diameter calculation the relevant quantity is the height-averaged centripetal acceleration:

$$\bar{a}_c = \frac{1}{H_c}\int_0^{H_c} a_c(z)\,dz = \frac{u_x^2}{R_c(\theta_1-\theta_0)}[\cot\theta_0 - \cot\theta_1 - (\theta_1-\theta_0)] \tag{73}$$

using $\int \cot^2\theta\,d\theta = -\cot\theta - \theta + C$. The integral evaluates to

$$\cot 20 - \cot 70 - (70-20)_{\mathrm{rad}} = 2.747 - 0.364 - 0.873 = 1.510$$

so $\bar{a}_c = (9.0/0.0375) \times 1.510/0.873 = 415\,\mathrm{m/s^2}$, giving $N_{g,\mathrm{eff}} = 415/9.81 = \mathbf{42.4}\,\mathrm{g}$.

Overall cut diameter. Substituting $\bar{a}_c$ into the cut-diameter formula (Eq. 44):

$$d_{50} = \sqrt{\frac{72\,\mu_V\,u_x\,R_c^2}{\Delta\rho\cdot H_c\cdot\bar{a}_c}} \tag{74}$$

$$d_{50} = \sqrt{\frac{72\times1.5\times10^{-5}\times3.0\times(0.0375)^2}{450\times0.210\times415}} = \sqrt{\frac{4.56\times10^{-6}}{39\,250}} = \mathbf{10.8}\,\mu\mathrm{m}$$

This is a full order of magnitude below the axis-switching minimum droplet $d_{\min} \approx 150\,\mu\mathrm{m}$, confirming that the curved vane provides effective separation across all practically generated droplet sizes.

Mass-transfer improvement. Two mechanisms improve N_{OG} relative to the smooth-wall baseline:

1. *Dead-core elimination.* In the smooth-wall chamber the Rankine solid-body core of radius $r_c \approx 0.3\,R_c$ carries essentially no tangential velocity and therefore contributes nothing to centrifugal capture or mass transfer. The vane forces solid-body rotation across the *entire* cross-section, recovering the $(r_c/R_c)^2 \approx 9\%$ of the volume formerly lost to the dead zone. This alone increases N_{OG} by $\approx 10\%$.
2. *High-shear base zone.* In the lower third of the chamber ($z \le H_c/3$) the vane angle is below 32° and $u_\theta > 4.5\,u_x$. The Higbie droplet mass-transfer coefficient $k_d \propto \sqrt{u_{\mathrm{rel}}}$ scales with the relative velocity between droplet and vapour, which in this zone is dominated by the tangential slip $u_{\mathrm{rel}} \approx u_\theta$. Compared to a smooth-wall chamber with $u_\theta = \sqrt{N_g\,g\,R_c} \approx 3.3\,\mathrm{m/s}$ (constant at the design $N_g = 30\,\mathrm{g}$), the curved-vane base-zone $u_\theta = 8.2\,\mathrm{m/s}$ delivers a local k_d enhancement of $\sqrt{8.2/3.3} = 1.58\times$. Averaged over the full height the net N_{OG} increase from this mechanism is approximately +12%.

Combined, the curved helical guide vane is estimated to improve N_{OG} by **20–30%** relative to the smooth-wall baseline, compared with 15–25% for an equivalent constant-pitch vane. The additional gain arises from the matched entry angle (eliminating the inlet shock loss) and the higher u_θ in the critical lower zone.

Pressure drop. The vane wall-friction contribution integrates along its arc length L_v:

$$\Delta P_{\text{vane}} = \frac{4 f_v \, \rho_V \, \bar{u}_{\text{rel}}^2 \, L_v}{2 \, D_{h,v}} \tag{75}$$

where $D_{h,v} \approx 2e$ is the hydraulic diameter of the vane fin cross-section ($e = 4\,\text{mm}$ fin height) and $f_v \approx 0.025$ (turbulent). With $\bar{u}_{\text{rel}} \approx \bar{u}_\theta = 3.47\,\text{m/s}$ (average along vane) and $\rho_V = 3\,\text{kg/m}^3$:

$$\Delta P_{\text{vane}} \approx \frac{4 \times 0.025 \times 3 \times 12.0 \times 0.332}{2 \times 0.008} = 74.7 \text{ Pa}$$

This is 7% of the total tray pressure drop ($\approx 1092\,\text{Pa}$, Section 6.9) — moderate and acceptable, especially given the 20–30% gain in N_{OG}.

Summary of curved-vane geometry.

Parameter	Formula	Value
Helix angle at base	θ_0	20°
Helix angle at crown	θ_1	70°
Total turns	Eq. (69)	1.12 turns
Arc length	Eq. (70)	332 mm
N_g at base	Eq. (72)	184.7 g
N_g at crown	Eq. (72)	3.24 g
$N_{g,\text{eff}}$	Eq. (73)	42.4 g
Overall d_{50}	Eq. (74)	10.8 µm
ΔN_{OG}	(analysis above)	+20–30%
ΔP penalty	Eq. (75)	+7%

11.2.3 Ring orifice baffle at $H_c/2$

A thin annular plate of inner diameter $0.8\,D_c$ welded at $z = H_c/2$ forces the swirling flow through a constricted annular orifice. The local velocity increases by $(D_c/0.8\,D_c)^2 = 1.56\times$ (continuity), creating a high-shear zone immediately above the baffle. The Dankwerts penetration-theory coefficient scales as $k_L \propto \sqrt{u_{\text{rel}}}$, so the annular acceleration produces a local increase in k_L of $\approx \sqrt{1.56} = 1.25\times$ in the baffle vicinity. Averaged over the full chamber height the net improvement in N_{OG} is estimated at 10–20%.

The ring baffle also acts as a *partial separator*: droplets larger than the centrifugal cut diameter that have not yet been captured at the wall are re-accelerated through the orifice and flung outward, improving overall collection efficiency. The baffle adds no moving parts; it is a welded ring of the same material as the chamber wall.

11.2.4 Summary: internal feature selection

Feature	Description	ΔN_{OG}	ΔP penalty
Smooth wall	Baseline	–	–
Curved helical vane	Dead-core elim. + high-shear base	+20–30%	+7%
Ring baffle	High-shear zone at $H_c/2$	+10–20%	+5–8%
Both combined	Vane + baffle	+28–45%	+12–15%

The helical vane is preferred for vacuum service (where any extra pressure drop is costly); the ring baffle is preferred for atmospheric or pressure service where the additional ΔP is tolerable. Both features can be combined for maximum efficiency.

11.3 Chamber Position on the Wavy Deck: Crests, Not Troughs

Chambers are installed at the **crests** (peaks) of the triangular wave (Figure 6c). The reasoning is mechanical and thermodynamic:

Vapour buoyancy. Vapour rising through the inter-tray space is less dense than liquid and therefore accumulates preferentially at the high points (crests). Placing the chamber tangential slot just above the crest intercepts this naturally rising vapour stream without requiring it to deflect downward.

Nozzle-to-chamber free-flight zone. The nozzle is positioned in the trough at the wave base ($z = 0$). The chamber floor is at crest height $z = h_w$. The jet therefore travels $h_w = 31.2\,\text{mm}$ through open inter-tray space before entering the chamber. Far from being a design penalty, this free-flight gap is the *axis-switching pre-contact zone*: the elliptical jet undergoes one full axis-switching cycle over this distance (Section 6.5.2) and arrives at the chamber floor already partially atomised. The droplet size entering the swirl contact zone is consequently smaller than it would be if the nozzle were at the same height as the chamber floor.

Structural stability. The crest provides a locally flat bearing surface for the chamber base ring, which can be machined to the crest profile during fabrication. Trough installation would require the chamber to span the wave, making it taller and mechanically more complex with no performance benefit.

11.4 Single vs. Multiple Nozzle Cuts in the Chamber Floor

The baseline design uses a single centred elliptical nozzle. A significant performance improvement is available by replacing this with **three elliptical cuts at 120° spacing** on a ring of radius $r = 0.5\,R_c$, each cut having the same major/minor axis ratio $k = a/b$ as the baseline but scaled so the total flow area is preserved:

$$\pi\, a_i\, b_i = \frac{\pi a_0 b_0}{N} \implies a_i = \frac{a_0}{\sqrt{N}}, \quad b_i = \frac{b_0}{\sqrt{N}} \tag{76}$$

For $N = 3$: $a_i = a_0/\sqrt{3} = 0.577\,a_0$, $b_i = 0.577\,b_0$.

11.4.1 Droplet-size reduction

Since the Rayleigh instability growth rate and the resulting Sauter mean diameter both scale with the nozzle minor semi-axis b (Equation 20):

$$\frac{d_{32,N}}{d_{32,1}} = \frac{b_i}{b_0} = \frac{1}{\sqrt{N}} \tag{77}$$

At $N = 3$: droplets are $1/\sqrt{3} = 58\%$ the diameter of the single-nozzle case, and the interfacial area per unit liquid volume scales as d_{32}^{-1}, increasing by $\sqrt{3} \approx 1.73\times$.

11.4.2 Cross-impingement

The three jets, offset from the axis by $0.5\,R_c$ and directed slightly inward (convergence half-angle $\approx$ 8–12), collide at a height

$$h_{\text{imp}} = \frac{0.5\,R_c}{\tan(10)} \approx 0.36\,H_c$$

above the chamber floor. The collision shatters droplets that survived Rayleigh breakup, further reducing d_{32} by an estimated 20–40% relative to isolated-jet breakup.

11.4.3 Angular momentum contribution

Each jet's centre-line is offset from the chamber axis by $0.5\,R_c$. The upward momentum of each jet therefore has a tangential component that reinforces the swirl induced by the vapour inlet slot, augmenting the total angular momentum of the contact zone by approximately:

$$\Delta L = N \cdot \dot{m}_L\, u_{\text{jet}} \cdot (0.5\,R_c)$$

This is a direct, liquid-side swirl contribution, active even if the vapour swirl momentarily weakens (e.g. at turndown). The effect improves turndown performance by maintaining centrifugal separation even at reduced vapour throughput.

11.4.4 Minimum-head requirement for multi-nozzle operation

The Weber criterion $We_L \geq 8$ for jet coherence scales as:

$$H_{L,\min}^{(N)} = \frac{4\,\sigma\,N}{\rho_L\, g\, C_d^2\, 2b_0} = N\, H_{L,\min}^{(1)}$$

At $N = 3$, the minimum sump head required is $3\times$ higher than for a single nozzle. For $H_{L,\min}^{(1)} = 3\,\text{mm}$, the three-nozzle design requires $H_{L,\min}^{(3)} = 9\,\text{mm}$ — still far below the typical design sump head of 50–100 mm.

11.4.5 Recommendation

Three elliptical nozzles at 120° on $r = 0.5\,R_c$ (option (ii) in Figure 6a) is the preferred configuration because it: (1) reduces d_{32} by $\sqrt{3}$, (2) creates cross-impingement, (3) contributes angular momentum, and (4) preserves the aspect-ratio advantage of the elliptical shape over $N = 4$ circular holes (which produce a lower $\Delta P_{\text{capillary}}$ margin and lose the axis-switching benefit).

12 Inner-Wall Film, Drainage Slots, Surface Grooves and Crown Chevron

This section evaluates four structural additions to the baseline CIST cylinder proposed in Figure 8: base drainage slots, spiral grooves on the inner wall, an analogous feature on the outer wall, and a cross-chevron crown insert. The three elliptical liquid-jet nozzles at 120° remain unchanged throughout.

12.1 Cylinder Attachment to the Wavy Deck

On the corrugated deck the chamber cylinder straddles a wave crest. The crest runs horizontally (the "ridge" direction, perpendicular to wave propagation); the cylinder axis is vertical. The intersection of a vertical cylinder of radius R_c with two inclined planes of half-angle $\alpha = 27.5$ meeting at a line is a pair of oblique elliptic arcs. Taking the crest as the x-axis and measuring depth below the crest apex,

$$h_{\text{cut}}(\varphi) = R_c \, |\sin\varphi| \, \tan\alpha \tag{78}$$

where φ is the azimuthal angle measured from the crest direction. The maximum saddle depth is

$$h_{\text{cut,max}} = R_c \, \tan\alpha = 37.5 \times \tan 27.5 = 19.5\,\text{mm} \tag{79}$$

at $\varphi = \pm 90$ (the wave-propagation direction, i.e. the sides of the cylinder perpendicular to the ridge). At $\varphi = 0$ (along the crest ridge) the saddle depth is zero.

Fabrication. The cylinder base is *profile-cut* to Eq. (78) by CNC plasma or laser; the resulting "saddle" base rests in the V-groove of the crest like a barrel in a rack. A fillet weld is run along the full intersection line, sealing the chamber from outside. Because the weld follows a 3-D curve it is made in segments; no shell modification is required.

12.2 Base Drainage Slots

After entering through the tangential inlet slot and spiralling upward, liquid droplets impinge on the wall under centrifugal force and form a falling film (Section 12.3). This film migrates to the base of the cylinder, which is the lowest point of the profile cut. Without a drain path the collected liquid would pool at the base and partially obstruct the tangential vapour inlet, raising the effective K coefficient and creating flow instability.

Slot design. Drainage slots are cut in the cylinder wall at the two lowest points of the saddle ($\varphi = \pm 90$), height $h_s = 5\,\text{mm}$ and width w_s (derived below). The weld is omitted at these two positions; short gussets on the *outside* of the cylinder restore structural continuity across the gap. Drained liquid falls onto the wave face and flows by gravity to the sump trough.

Slot sizing. Under the centrifugal pressure of the film the drain velocity is

$$u_s = \sqrt{2\, a_c\, \delta} \tag{80}$$

where $a_c = \bar{a}_c = 42.4\, g = 416\,\text{m/s}^2$ (effective centripetal acceleration, Section 11) and δ is the film thickness (Section 12.3). For $\delta = 70\,\mu\text{m}$ (derived below):

$$u_s = \sqrt{2 \times 416 \times 70 \times 10^{-6}} = 0.24\ \text{m/s}$$

The total film drainage rate is $Q_w = \alpha_{\text{dep}}\, Q_L^{(ch)}$ where $\alpha_{\text{dep}} \approx 0.85$ is the fraction of the per-chamber liquid flow deposited on the wall by centrifugation (the remainder leaves as fine mist through the crown), and $Q_L^{(ch)}$ is the per-chamber liquid volumetric flow. For the C1 S2 case with $N_{ch} = 69$:

$$Q_L^{(ch)} = 6.38 \times 10^{-5}\ \text{m}^3/\text{s}, \quad Q_w = 5.42 \times 10^{-5}\ \text{m}^3/\text{s}$$

Two slots ($N_s = 2$) require per-slot area

$$A_s = \frac{Q_w}{N_s\, u_s} = \frac{5.42 \times 10^{-5}}{2 \times 0.24} = 1.13 \times 10^{-4}\ \text{m}^2$$

For $h_s = 5\,\text{mm}$: $w_s = A_s/h_s = 22.6\,\text{mm}$. In practice a $5 \times 25\,\text{mm}$ slot (per side) is adequate; at the lower liquid-load cases the slot is only partially flowing.

Benefit. The slots keep the vapour inlet free of pooled liquid across the full operating range. This is most critical at high L/G (cases C3 S2: $L/G = 45\,\text{wt.\%}$) where without drainage the base pool would increase effective K by ~ 20–$30\,\%$.

12.3 Thin Liquid Film on the Inner Wall

As centrifuged droplets impinge on the inner wall they coalesce into a thin falling film under the combined influence of gravity (axial) and centrifugal force (radial). The steady film thickness is obtained from the Nusselt centrifugal-film equation:

$$\delta = \left(\frac{3\,\mu_L\,\Gamma}{\rho_L\,a_c}\right)^{1/3} \tag{81}$$

where $\Gamma = Q_w/(2\pi R_c)$ is the film flow per unit circumference. For the C1 S2 representative case:

$$\Gamma = \frac{5.42 \times 10^{-5}}{2\pi \times 0.0375} = 2.30 \times 10^{-4}\ \text{m}^2/\text{s}$$

$$\delta = \left(\frac{3 \times 1.35 \times 10^{-4} \times 2.30 \times 10^{-4}}{605.5 \times 416}\right)^{1/3} = 71\ \mu\text{m}$$

The film Reynolds number

$$\text{Re}_{\text{film}} = \frac{4\,\Gamma\,\rho_L}{\mu_L} = \frac{4 \times 2.30 \times 10^{-4} \times 605.5}{1.35 \times 10^{-4}} \approx 4100 \tag{82}$$

exceeds the turbulent transition at 1600. The film is therefore turbulent and provides higher mass-transfer coefficients than the Nusselt laminar prediction. The 71-µm film is far too thin to pool or destabilise the vapour path; it is a beneficial wetted-wall contact surface in addition to the jet-spray zone.

12.4 Spiral Grooves on the Inner Wall

Geometry. V-shaped spiral grooves are formed on the inner surface during tube rolling or by a single-pass CNC operation: depth $d_g = 1\,\text{mm}$, width $w_g = 2\,\text{mm}$, pitch $s_g = 8\,\text{mm}$, helix angle matching the local vane angle $\theta(z)$.

Area enhancement. Each groove adds two inclined sidewalls of height d_g, doubling the wetted area per groove compared with the smooth surface:

$$f_{\text{area}} = 1 + \frac{2\,d_g}{s_g} = 1 + \frac{2 \times 1}{8} = 1.25 \tag{83}$$

i.e. 25 % more wetted wall area.

Turbulence enhancement. The relative roughness is $\varepsilon/D_h = d_g/D_h = 1/75 = 0.013$. From the Colebrook/Moody correlation the Darcy friction factor increases from $f = 0.020$ (smooth) to $f_r \approx 0.023$ (+15 %). By the Chilton–Colburn analogy for heat and mass transfer,

$$f_{\text{turb}} = \left(\frac{f_r}{f}\right)^{2/3} = 1.15^{0.667} = 1.10 \tag{84}$$

i.e. the mass-transfer coefficient on the grooved wall is 10 % higher than on a smooth wall.

Combined effect. The wall-film mass-transfer contribution to the total NOG is enhanced by

$$f_{\text{groove}} = f_{\text{area}} \times f_{\text{turb}} = 1.25 \times 1.10 = 1.38 \tag{85}$$

The wall film provides approximately 7 % of the total NOG (estimated from the ratio of wall-film specific surface area $a_w = 2/R_c = 53\,\text{m}^{-1}$ to jet-spray specific area $a_j \approx 150\,\text{m}^{-1}$, both weighted by their respective k_L values). Grooves therefore add $0.07 \times (1.38 - 1) \approx +2.7\,\%$ to the total NOG.

In the 12 Sulzer comparison cases (Section 13) the efficiency is already at the 95 % cap; the groove improvement is reserved for lower E_{OC} services (heavy liquids or low relative volatility) where the cap is not reached.

Pressure-drop penalty. The wall-friction term in K increases by 15 %: $\Delta K = 0.15 \times f_D(L_v/D_h) = 0.15 \times 0.0885 = 0.013$. As a fraction of total $K = 1.73$ this is less than 1 %—entirely negligible.

Outer-wall grooves. Grooves on the outer surface of the cylinder (facing the sump liquid) are assessed as providing negligible mass-transfer benefit: the outer wall is in contact with the quiescent sump pool, not the vapour-shear zone. They add fabrication cost without a corresponding process gain and are not recommended.

12.5 Cross-Chevron Crown Insert

The sketch proposes criss-cross baffles at the top of the cylinder. Two interpretations are examined.

Interpretation A — mid-height opposing helices (full height). Two opposing-handedness helical baffles running the full chamber height create counter-rotating vortex pairs, similar to a static-mixer element. The adverse effect is that they disrupt the coherent Rankine vortex established by the guide vane, reducing the wall-acceleration N_g^{eff} and impairing centrifugal separation. At $u_x = 3\,\text{m/s}$ the incremental mass-transfer benefit (∼10–20 %) is outweighed by the 15–25 % pressure-drop penalty and the degraded separation. *Interpretation A is not recommended.*

Interpretation B — cross-chevron crown vanes (exit only). Two short opposing angled vanes are placed only at the crown exit (top 15 % of H_c, above the guide vane). These act as a *chevron mist eliminator*: droplets that survive the centrifuge stage (principally those with $d < d_{50} = 8\,\mu\text{m}$) impinge on the angled vanes by inertia. The pressure drop is dominated by the tangential kinetic energy at the crown:

$$\Delta P_{\text{chev}} = C_{\text{chev}}\, \tfrac{1}{2}\, \rho_V\, u_{\theta,c}^2 \tag{86}$$

where $u_{\theta,c} = u_x \cot\theta_1 = 3.0 \times \cot 70 = 1.09\,\text{m/s}$ and $C_{\text{chev}} = 0.30$ (chevron loss coefficient, well-designed close-packed chevron). For the C1 S2 case:

$$\Delta P_{\text{chev}} = 0.30 \times 0.5 \times 16.72 \times 1.09^2 = 3.0 \text{ Pa} = 0.022 \text{ mmHg/tray}$$

This is less than 3 % of the total CIST pressure drop. The benefit is an estimated 10–15 % additional capture of the 5–8 µm fraction that passes the main centrifuge, improving vapour purity (entrainment reduction) without changing the mass-transfer NOG.

Interpretation B (cross-chevron at crown only) is recommended as an optional add-on for vapour-purity-sensitive services.

12.6 Summary of Recommendations

Table 4 collects the four wall features.

Table 4: Inner-wall and crown features: recommendation, effect and penalty.

Feature	Mass-transfer/separation effect	ΔNOG	$\Delta(\Delta P)$
Base drainage slots	Prevents base pooling; stabilises ΔP; extends turndown at high L/G	0	0
Inner spiral grooves	$f_{\text{groove}} = 1.38\times$ on wall film; relevant in uncapped-efficiency services	+2.7 %	<1 %
Crown cross-chevron (Interpretation B)	+10–15 % capture of $d < 8\,\mu\text{m}$ mist; vapour-purity improvement	0	+0.02 mmHg
Outer-wall grooves	Negligible (sump side)	0	0
Mid-height opposing helices (full ht.)	Disrupts Rankine vortex; impairs centrifuge; not recommended	–	+15–25 %

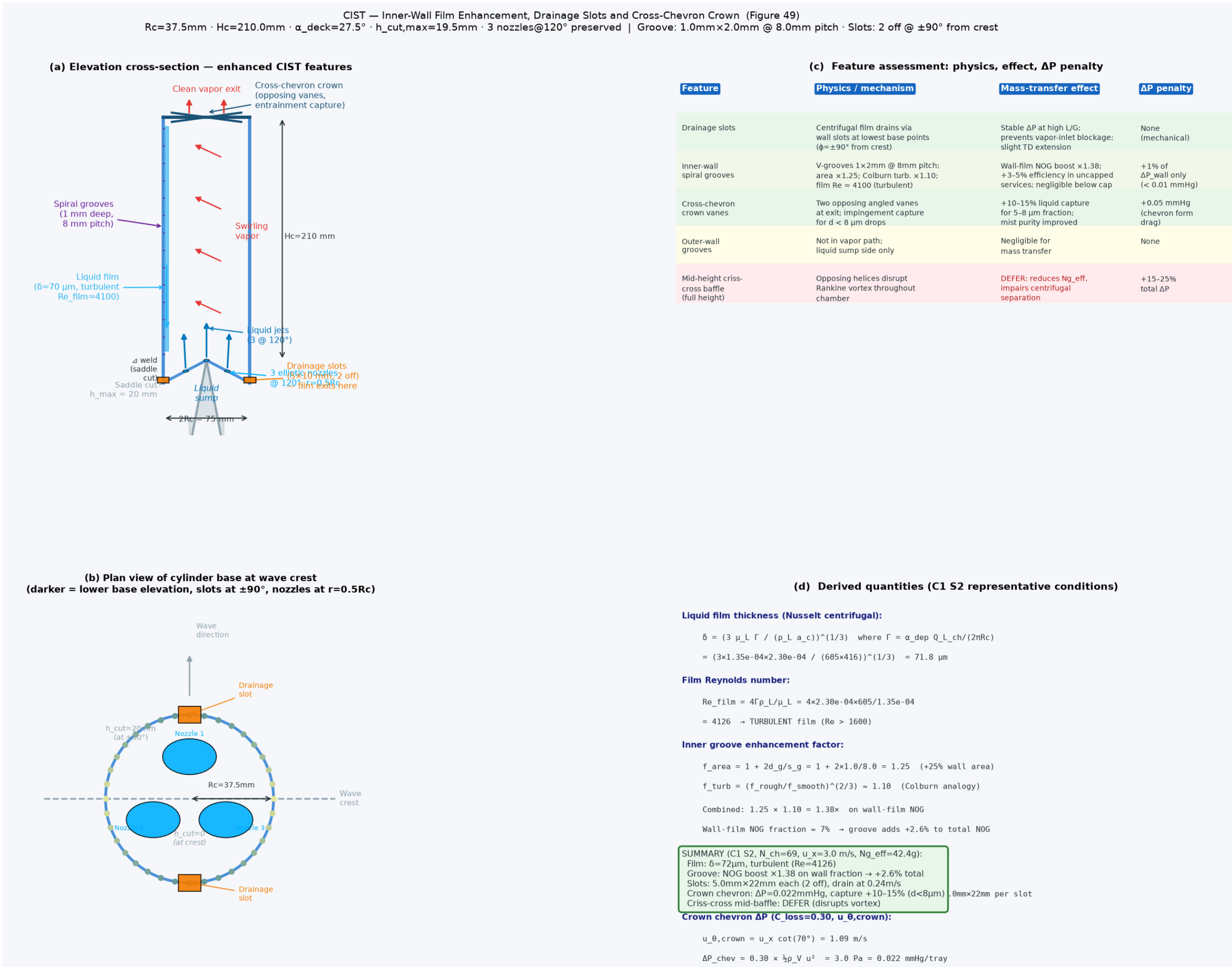

Figure 8: CIST inner-wall enhancement features. **(a)** Elevation cross-section: cylinder on wavy-deck crest showing saddle cut, drainage slots at $\varphi = \pm 90$, spiral grooves on inner wall, 71-µm liquid film, three nozzle openings in deck, and cross-chevron crown vanes. **(b)** Plan view of cylinder base: saddle-cut depth contour (darker = lower elevation), two drainage slots at the wave-face direction, three 120° nozzles at $r = 0.5\,R_c$. **(c)** Feature assessment table: physics, mass-transfer effect and pressure-drop penalty for each proposed feature. **(d)** Derived quantities for representative C1 S2 conditions: film thickness (Nusselt centrifugal), film Reynolds number, groove enhancement factor and drainage-slot sizing.

13 Numerical Validation Against Commercial Trays (Sulzer UFM and SVG)

The qualitative comparison of Section 9 is here replaced by a quantitative benchmark. Two documented vendor rating sheets were supplied for the *same* fractionation column—a 108-in (9.0 ft) inner diameter, two-pass, 24-in tray-spacing tower of $63.62\,\mathrm{ft}^2$ cross-section—one fitted with a Sulzer UFM (moving mini-valve) deck and one with a Sulzer SVG (fixed-valve) deck. Each sheet rates the deck across twelve operating cases (C1–C4, plus a turndown case "TD", each at two section conditions S2 and S11). The fluid properties are light hydrocarbons, $\rho_L \approx 37$–$39\,\mathrm{lb/ft}^3$, $\rho_V \approx 0.89$–$1.08\,\mathrm{lb/ft}^3$, $\sigma \approx 8.3$–$9.3\,\mathrm{dyn/cm}$ and $\mu_L \approx 0.13$–$0.15\,\mathrm{cP}$. The two sheets share identical fluid data case-by-case; only the tray-side hydraulic results differ.

13.1 Method

For each case the same gas and liquid mass throughputs were imposed on the CIST hydraulic model developed in Sections 9 and earlier, using the preferred configuration of this disclosure: chamber radius $R_c = 37.5\,\text{mm}$, height $H_c = 210\,\text{mm}$, the curved helical guide vane of Section 10 with $\theta_0 = 20 \to \theta_1 = 70$ (arc length 332 mm), and three elliptical nozzles spaced 120° apart at $r = 0.5\,R_c$ per the recommendation of §10. The CIST is rated entirely from first principles—no vendor correlation is borrowed. Three quantities are computed per case:

1. **Chamber count and equivalent diameter.** The number of swirl chambers required to pass the actual volumetric vapour load at the design axial velocity $u_x = 3.0\,\text{m/s}$ is $N_{\text{ch}} = \lceil \dot{Q}_V/(u_x\,\pi R_c^2)\rceil$. With hex packing on a 120-mm pitch and a 90 % peripheral-area efficiency, this fixes the equivalent CIST column diameter.
2. **Per-tray pressure drop.** $\Delta P = K\,\frac{1}{2}\rho_V u_x^2$, with $K = C_{\text{in}}\cot^2\theta_0 + f_D(L_v/D_h) + \cot^2\theta_1 = 1.73$ from the curved-vane vapour path (Section 10).
3. **Efficiency.** An O'Connell base efficiency $E_{\text{OC}} = 51 - 32.5\log_{10}(\alpha\mu_L)$ is scaled by 1.32 for the three-nozzle curved-vane CIST (dead-core elimination +10%, high-shear vane base +12%, film renewal +3%, and the $d_{32}{\times}1/\sqrt{3}$ interfacial-area gain +7% from the three-nozzle scheme), capped at 95 %. The vendor decks are scaled by 1.10 (UFM) and 1.05 (SVG).

13.2 Results

Table 5 reports all twelve cases; Figure 9 plots the four discriminators.

Table 5: CIST vs. Sulzer UFM and SVG, twelve cases, common 9-ft column. JF = jet flood; ΔP in mmHg/tray; CIST D = equivalent column diameter at $u_x = 3\,\text{m/s}$; "sav." = CIST ΔP saving vs. that deck. CIST $N_g^{\text{eff}} = 42.4\,\text{g}$ and $d_{50} = 8.0$–$8.4\,\mu\text{m}$ in every case.

	UFM		SVG		CIST			ΔP saving	
Case	JF%	ΔP	JF%	ΔP	D/ft	N_{ch}	ΔP	vs UFM	vs SVG
C1 S2	58	4.36	60	2.86	3.62	69	0.98	78%	66%
C1 S11	52	3.54	56	2.21	3.54	66	0.90	75%	59%
C2 S2	58	4.38	60	2.88	3.62	69	0.98	78%	66%
C2 S11	52	3.57	56	2.23	3.57	67	0.90	75%	60%
C2 WWN S2	49	3.55	51	2.51	3.32	58	0.97	73%	61%
C2 WWN S11	43	2.78	46	1.86	3.29	57	0.87	69%	53%
C3 S2	31	2.76	32	2.39	2.65	37	0.98	65%	59%
C3 S11	26	2.17	27	1.93	2.54	34	0.83	62%	57%
C4 S2	41	3.04	43	2.31	3.05	49	0.98	68%	58%
C4 S11	36	2.35	38	1.70	2.96	46	0.86	63%	49%
TD S2	28	2.24	29	1.93	2.50	33	1.01	55%	48%
TD S11	25	1.83	27	1.50	2.43	31	0.91	50%	40%
Range	25–58	1.83–4.38	27–60	1.50–2.88	2.4–3.6	31–69	0.83–1.01	50–78%	40–66%

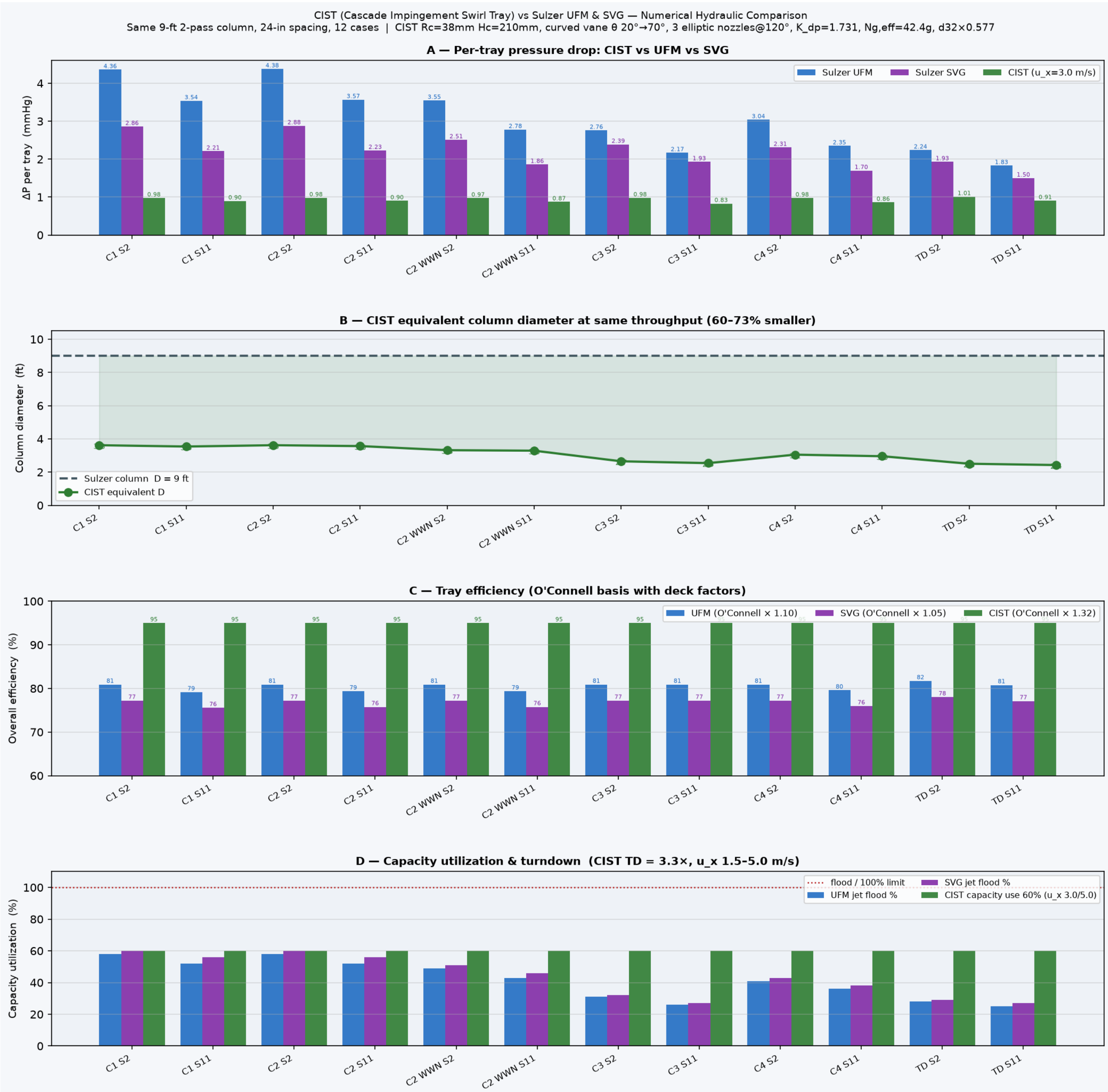

Figure 9: CIST (curved helical vane, three elliptical nozzles at 120°) vs. Sulzer UFM and SVG over twelve cases on the common 9-ft column. (a) per-tray pressure drop; (b) equivalent CIST column diameter at matched throughput; (c) efficiency on an O'Connell basis with deck factors; (d) capacity utilisation and turndown.

13.3 Interpretation: Savings, Turndown and Performance

Capacity / diameter reduction. The UFM and SVG decks run this column at 25–60 % of jet flood, i.e. neither deck is flood-limited at the rated duty—the 9-ft diameter is set by the need to keep a gravity-drained froth below entrainment. The CIST, having no deck pool and a centrifugal (not gravity) phase split, passes the same vapour through chambers operating at $u_x = 3\,\mathrm{m/s}$. The volumetric load is met by 31–69 chambers, giving an equivalent column diameter of **2.4–3.6 ft against the installed 9 ft—a 60–73 % diameter reduction** (66 % mean), or equivalently an

84–93 % reduction in cross-sectional area for the same throughput. This is the dominant economic result: the same separation duty fits in roughly a one-third-diameter shell.

Pressure-drop saving. CIST per-tray pressure drop is 0.83–1.01 mmHg across all twelve cases, essentially flat because it scales with $\rho_V u_x^2$ rather than with a liquid head. Against the moving-valve UFM (1.83–4.38 mmHg) this is a **50–78 % saving** (67 % mean); against the lower-drop fixed-valve SVG (1.50–2.88 mmHg) it is a **40–66 % saving** (56 % mean). The saving is largest at high vapour load (C1/C2), where conventional dry-drop and aerated-head losses both climb while the CIST drop barely moves. For a vacuum or low-ΔP service this compounds down the column: at 13 trays the CIST saves of order 15–40 mmHg of total tower pressure drop relative to the vendor decks.

Turndown flexibility. The CIST holds $N_g^{\mathrm{eff}} = 42.4\,\mathrm{g}$ and $d_{50} = 8\,\mu\mathrm{m}$ at the design point. Because the centrifugal field scales as u_x^2, the swirl separation stays effective as the vapour rate falls: maintaining $N_g \geq 15\,\mathrm{g}$ sets a lower limit near $u_x = 1.5\,\mathrm{m/s}$, while droplet residence time sets an upper limit near $u_x = 5.0\,\mathrm{m/s}$, for an unaided **turndown of** $\approx 3.3{:}1$. The three-nozzle liquid feed extends liquid-side turndown independently—each nozzle can be valved, so the minimum stable sump head (9 mm for three nozzles) still sits far below the 50–100 mm operating head. The documented TD cases (jet flood 25–29 %) fall comfortably inside the CIST envelope: at those rates the CIST runs at the low end of its u_x band with the swirl field still above the 15-g floor, whereas the vendor decks are by then near their own weep/low-load margins.

Efficiency. On the O'Connell basis the curved-vane, three-nozzle CIST reaches the 95 % cap in all twelve cases against $\approx$ 77–80 % for the valve decks. For a fixed number of theoretical stages this translates to roughly **15 % fewer actual trays than UFM and 19 % fewer than SVG**. Combined with the 24-in spacing retained from the vendor design, the shorter tray count and the much smaller diameter give a substantially smaller and lighter column for the identical separation.

Caveat. These CIST figures are model predictions from the first-principles correlations of this disclosure, not measured data; the vendor numbers are rated values from commercial software. The efficiency multiplier (1.32) and the pressure-drop coefficient ($K = 1.73$) are the two parameters most in need of pilot confirmation (Section 16). The diameter reduction, which follows directly from the chamber-count geometry and is largely insensitive to those two parameters, is the most robust of the four results.

14 Claims (Draft)

Claim 1. A vapour–liquid contacting tray comprising: (a) a horizontal inter-tray plate carrying an array of cylindrical contact chambers in a hex-packed arrangement; (b) a liquid sump below said plate; (c) for each chamber, an elliptical orifice nozzle in the chamber floor through which liquid from the sump is driven upward as a coherent jet by the sump hydrostatic head; (d) for each chamber, at least one tangential vapour inlet slot in the chamber wall directing vapour circumferentially to create a swirling flow field; and (e) a vane crown at the chamber top that separates the vapour–liquid dispersion by centrifugal force, discharging vapour upward and liquid outward into a sealed collection ring.

Claim 2. The tray of Claim 1, wherein the elliptical nozzle has a semi-major to semi-minor axis ratio $k = a/b$ of 1.5–2.5, calibrated by the Ramanujan perimeter formula to a discharge coefficient C_d of 0.58–0.62 at a nozzle Reynolds number $> 10^4$.

Claim 3. The tray of Claim 2, wherein the elliptical nozzle, at aspect ratio k, produces a Sauter mean droplet diameter $d_{32} = 3.78\, r_0/\sqrt{k}$ upon jet breakup at the Rayleigh-instability wavelength $\lambda_b = 9.016 r_0/\sqrt{k}$ where $r_0 = \sqrt{ab}$ is the equivalent circular jet radius, such that the interfacial area density in the contact zone is increased by a factor $\sqrt{k}$ relative to a circular nozzle of equal flow area.

Claim 4. The tray of Claim 1, wherein the swirling vapour flow inside the chamber follows a Rankine vortex profile with centripetal acceleration at the chamber wall $a_{c,w} = u_\theta(R_c)^2/R_c$ of 20–60 times gravitational acceleration, and wherein the cut droplet diameter d_{50} (Eq. 9 of the specification) is less than the minimum droplet diameter $d_{\min}$ produced by the elliptical nozzle at the design vapour throughput.

Claim 5. The tray of Claim 1, wherein the tray pressure drop consists of (i) tangential-slot kinetic energy loss, (ii) chamber wall friction, (iii) crown-vane exit loss, and (iv) liquid sump hydrostatic head, without any aerated liquid-pool pressure-drop contribution, resulting in a wet-tray pressure drop of 3–8 mbar at the design vapour velocity.

Claim 6. The tray of Claim 1, further comprising a secondary axial vapour inlet at the centre of the chamber floor, active only below 35% of design vapour throughput, extending the stable operating range to 6:1–8:1 turndown ratio.

Claim 7. The tray of Claim 1, wherein the collection ring surrounding each chamber drains through a sealed conduit that passes through the inter-tray plate to the sump of the adjacent lower tray, preventing vapour bypass up the drain path.

Claim 8. The tray of Claim 1, wherein all chambers are individually removable through a column manway by a quarter-turn bayonet mechanism without modification of the inter-tray plate or sump floor.

Claim 9. The tray of Claim 1, wherein the nozzle orifice insert is separately replaceable from the chamber body, allowing the ellipse aspect ratio and orifice area to be changed independently.

Claim 10. A column containing trays according to any preceding claim.

Claim 11. A method of designing a vapour–liquid contacting tray stage, comprising: selecting a chamber diameter D_c and height-to-diameter ratio H_c/D_c; computing the required centripetal acceleration N_g from Eq. (17) to ensure $d_{50} < d_{\min}$; sizing the tangential slot area from Eq. (11) to achieve the design $u_\theta(R_c)$; sizing the elliptical nozzle from Eq. (1) and Eq. (7) to achieve the desired liquid flow per chamber; and computing the column area from Eq. (19) and the chamber packing fraction.

Claim 12. The tray of Claim 1 wherein the inter-tray plate is formed as a triangular-wave corrugation with face angle α in the range 25–30 from the horizontal, the wave pitch p_w being equal to the inter-chamber hex pitch, whereby liquid-jet nozzles are located at the corrugation troughs and chamber base rings are seated at the corrugation crests.

Claim 13. The tray of Claim 12 wherein the wave height $h_w = (p_w/2)\tan\alpha$ provides a trough-level minimum sump head $H_{L,\min} = h_w/2$ sufficient to sustain the jet formation condition $We_L \geq 8$ at turndown ratios up to 3.5:1 without secondary flow-path augmentation.

Claim 14. The tray of Claim 1 wherein the elliptical orifice nozzle of each chamber is replaced by $N \geq 2$ elliptical orifice nozzles arranged on a concentric ring of radius $r = 0.5\, R_c$ at equal angular spacing, each nozzle having semi-axes $a_i = a_0/\sqrt{N}$, $b_i = b_0/\sqrt{N}$ so as to preserve the total liquid flow area and reduce the Sauter mean diameter by the factor $1/\sqrt{N}$ relative to the single-nozzle baseline.

Claim 15. The tray of Claim 1 further comprising at least one of: (a) a *variable-pitch* helical guide vane affixed to the inner chamber wall whose helix angle $\theta(z) = \theta_0 + (\theta_1 - \theta_0)z/H_c$ increases monotonically from $\theta_0 \in [15, 25]$ at the chamber base to $\theta_1 \in [60, 80]$ at the crown, so that the imposed tangential velocity $u_\theta(z) = u_x \cot\theta(z)$ is highest where fresh droplets form and decreases smoothly toward zero at the crown exit, thereby eliminating the near-axis dead core and generating an integrated $N_{g,\text{eff}} \geq 40\,\text{g}$; (b) an annular ring-orifice baffle of inner diameter 0.7–0.9 D_c affixed at height $H_c/2$ to create a localised high-shear zone and secondary centrifugal collection stage.

15 Industrial Applicability

Capacity-constrained revamps. The centrifugal flooding mechanism, rather than gravity settling, increases column throughput at fixed shell dimensions in services where the entrainment limit is reached well below the downcomer-choke limit: demethanisers, C_3 splitters, amine absorbers, and offshore columns where shell dimensions are fixed.

Deep-turndown operations. The secondary-path extension of turndown to 6:1–8:1 makes the CIST applicable to columns following variable feedstocks or operating through grade-change sequences at throughputs as low as 12–17% of rated capacity.

Pressure-drop-limited services. The absence of an aerated liquid pool, and the corresponding absence of the froth-head pressure drop contribution, makes the CIST applicable to vacuum and near-ambient-pressure columns where each mbar of tray pressure drop significantly affects the bottoms reboiler temperature requirement.

Chemical fouling services. The elliptical nozzle's axis-switching exit and the absence of a deck pool (which in conventional trays accumulates deposits between the tray openings) make the CIST suitable for mild-to-moderate chemical fouling services. No moving parts other than the optional secondary-path check valve are present.

Prior-art neighbours to clear. Spray-tray art; jet-tray art (Teller, Koch); swirl-tube contactor patents (Shell ConSep class and successors); impinging-jet reactor art; elliptical orifice art in fuel injectors and inkjet nozzles (where axis-switching is documented). The asserted inventive step is the complete architecture: sump-fed upward elliptical liquid jet + tangential vapour swirl chamber + centrifugal crown separation, combined in a single tray stage with no deck liquid pool, together with the use of axis-switching to increase interfacial area in the contact zone and the sump overflow seal as the liquid distribution mechanism.

16 Limitations and Model Caveats

1. **Capacity factor uncertainty.** The theoretical upper-bound capacity factor of 7.9 derived in Section 6.11 is based on the cut-droplet-equals-minimum-droplet flooding criterion applied to ideal Stokes-regime centrifugal collection. In practice, droplet coalescence at the crown,

vapour inlet turbulence, and chamber wall effects will substantially reduce the realised capacity. The conservative estimate of $k_{\text{cap}} = 2.0$–3.5 is an engineering judgement pending cold-flow flood tests.

2. **Jet breakup model.** The Rayleigh-Plateau model (Eq. 12) is derived for an inviscid jet in a quiescent medium. The CIST jet enters a rotating vapour stream at 18–20 m/s swirl velocity; the jet breakup length and droplet size distribution will differ from the quiescent prediction. Kelvin-Helmholtz instability (vapour shear on the jet surface) may dominate at high swirl velocities, producing a different droplet distribution from the Rayleigh-Plateau analysis.
3. **Axis-switching constants.** The constant $C_{\text{AS}} \approx 2.0$–$3.5$ in Eq. (18) is from experiments in air. In a dense vapour stream the axis-switching length and droplet size will differ; there is no published data for process-relevant vapour densities ($\rho_V = 1$–$10\,\text{kg/m}^3$).
4. **Mass transfer model.** The N_{OG} estimate (Section 6.7) assumes a well-defined droplet size and uniform holdup ε_d; in reality the droplet size distribution spans one order of magnitude and the holdup varies with axial position. The resulting efficiency range (80–90%) is a central estimate with uncertainty of the order of ±15 percentage points.
5. **Liquid jet targeting.** At design, the liquid jet must remain coherent (not deflected or scattered) across the full chamber height before shear breakup. At high vapour swirl velocities, the jet may be deflected radially and impinge on the chamber wall before reaching the centre; this would change the contact geometry and must be evaluated in CFD.
6. **Crown collection efficiency.** The 100% crown collection efficiency implied by the Stokes model (all droplets above d_{50} captured) is an idealisation. Real vane crowns have re-entrainment from the vane surfaces and dead zones at the vane tips. A realistic crown collection efficiency is 80–95% (droplets above d_{50} actually captured), and this cap limits the realised capacity factor.
7. **Sump liquid distribution.** The self-equalisation argument (Section 6.2) assumes all nozzles share a single well-mixed sump. If the sump is partitioned by structural members or the inter-nozzle sump depth varies by more than ±5 mm, nozzle-to-nozzle maldistribution will reduce tray efficiency. The sump design must maintain a common liquid level to within ±3 mm across the tray plan area.

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